Renormalization Group Study of Constrained Hamiltonians

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1 Introduction

Renormalization Group methods can be used to explore the behaviour of physical systems near a critical point, which corresponds to a fixed point under the renormalization group (RG) flow. These fixed points are characterised by critical exponents that express the behaviour of observables as criticality is approached. For example, the exponent ν is defined by $\xi = a^{-\nu}$, where ξ is the correlation length of the system and a is its (reduced) temperature. It is well known that many systems share the same critical exponents, that is, belong to the same universality class.

The universality class seem to be closely related to the symmetries of the physical system. For example, the Ising universality class corresponds to $\mathbb{Z}_2$ symmetry. However, it has been shown recently that adding a constraint to the system that preserves its symmetry may change its universality class [1]. These constraints end up changing the fixed point in the RG flow.

In this essay, our aim is to understand how constrained Hamiltonians give rise to new universality classes. We will start with a revision of statistical field theory based on [2, 3, 4, 5, 6]. We begin with a recap of statistical physics, then move on to functional methods, discussing mean field theory and fluctuations. Then we will briefly describe the renormalization group theory, and finally present a review of RG of constrained Hamiltonians based on [1].

1.1 Statistical Physics

In modern formulations of statistical mechanics, all follows from one postulate, the principle of maximum entropy:

Postulate. *A system in equilibrium maximises its Gibbs entropy, subject to constraints.*

The Gibbs entropy is given by

$$S = -k_B \sum_\theta p(\theta) \log p(\theta)$$

where $\sum_\theta$ is a sum over all possible microstates, $p(\theta)$ is the probability of a microstate labelled by θ and k_B is the Boltzmann constant.

In general, we will be concerned with a system of fixed volume and particle number, immersed in a heat bath such that the internal energy average E is fixed. The principle of maximum entropy is applied by maximising the Gibbs entropy with one Lagrange multiplier for the energy constraint and another for the probabilities summing to a constant which we take as 1:

$$\frac{\partial}{\partial p(\theta)}\Big(S - \lambda_1 \sum_\theta p(\theta) - \lambda_2 \sum_\theta p(\theta)E(\theta)\Big) = 0$$
$$\Rightarrow -k_B(\log p(\theta)) + 1) - \lambda_1 - \lambda_2 E(\theta) = 0 \Rightarrow p(\theta) = \frac{e^{-\beta E(\theta)}}{Z},$$

where $\beta = \lambda_2/k_B$ and Z is called the partition function. This is known as the canonical ensemble. In order for the distribution to be normalised, we must have:

$$Z = \sum_\theta e^{-\beta E(\theta)}. \tag{1}$$

The parameter β in (1) is related to the thermodynamic temperature. The 1st law of thermodynamics is

$$dE = TdS - pdV + \mu dN.$$

Differentiating with respect to E at fixed V and N, we get:

$$\frac{\partial S}{\partial E}\Big|_{V,N} = \frac{1}{T},$$

but

$$S = -k_B \sum_\theta p(\theta)(-\beta E(\theta) - \log Z)$$
$$= k_B \beta E + k_B \log Z$$

where E is the average energy. Taking the derivative with respect to E we find:

$$\frac{\partial S}{\partial E} = k_B \beta = \frac{1}{T} \Rightarrow \beta = \frac{1}{k_B T}.$$

A more useful object to use than E is the Helmholtz free energy:

$$F(T, N, V) = E - TS = -k_B T \log Z$$
$$\Rightarrow dF = dE - TdS - SdT = -SdT - pdV + \mu dN$$

such that, in the canonical ensemble, F is minimised at equilibrium.

1.2 Building Blocks

We refer to anything between a microstate and a macrostate as a mesostate. A mesostate description can be obtained from a microstate description using a coarse-graining procedure. This consists in averaging the system over a finite number of microscopic degrees of freedom.

For example, a coarse-grained field $\Psi(\mathbf{r})$ can describe the average field near each location $\mathbf{r}$. This means that a single configuration $\Psi(\mathbf{r})$ corresponds to a large number of microstates with the same coarse-grained appearance. We refer to these states collectively as $\{\theta|\Psi\}$. The coarse-grained field is also known as the order parameter.

We can then define a restricted partition function

$$Z[\Psi] = \sum_{\{\theta|\Psi\}} e^{-\beta E(\theta)} \equiv e^{-\beta F[\Psi]}$$

where I have defined the effective free energy $F[\Psi]$. Note that

$$F[\Psi] = E[\Psi] - TS[\Psi] = -k_B T \log Z[\Psi]$$

where

$$E[\Psi] = \sum_{\{\theta|\Psi\}} p(\theta) E(\theta)$$

$$S[\Psi] = k_B T \sum_{\{\theta|\Psi\}} p(\theta) \log p(\theta)$$

The probability for a coarse-grained field configuration $\Psi(\mathbf{r})$ is then

$$p[\Psi] = \frac{1}{Z_{TOT}} \sum_{\{\theta|\Psi\}} e^{-\beta E(\theta)} = \frac{e^{-\beta F[\Psi]}}{Z_{TOT}} \tag{2}$$

and the partition function can be written as a functional integral over field configurations:

$$Z_{TOT} = \sum_{\Psi} e^{-\beta F[\Psi]} = \int D\Psi e^{-\beta F[\Psi]} \tag{3}$$

From the partition function, we can derive the the Helmholtz free energy

$$F = -k_B T \log Z_{TOT}.$$

Any physical system will be described in terms of a partition functional (3), from which probabilities can be calculated as in (2).

It is in our interest then to look for the most general free energy, capable of describing a large array of systems and phenomena. To write it down we use a phenomenological approach: we list some desirable properties and search for the most general function that satisfies them. The result is the Landau-Ginzburg theory.

1.3 The Landau-Ginzburg Free Energy

We consider as our order parameter an N-component field, $\phi : \mathbb{R}^d \to \mathbb{R}^N$ where d is the spatial dimension. As some examples, $N = 1$ describes binary fluids and uniaxial magnets (also known as the Ising model), $N = 2$ describes superfluids and planar magnets, $N = 3$ describes classical magnets. The criteria that the free energy should satisfy are:

(i) Locality and uniformity: the probability of a field configuration $p[\phi(\mathbf{r}]$ should be a product of contributions from different locations - in a magnet for instance the spins only affect their close neighbours. This translates to F being a sum (or integral) over all space:

$$F[\phi(\mathbf{r})] = \int d\mathbf{r} f(\mathbf{r}, \phi(\mathbf{r}), \nabla\phi(\mathbf{r}), ...)$$

and uniformity allows us to drop the $\mathbf{r}$ dependency.

(ii) Analiticity: As coarse-graining averages out many microscopic degrees of freedom, we assume that any singularities in the microscopic description are washed away, and that $f(\phi(\mathbf{r}))$ can be written as a Taylor series at any point.

(iii) Symmetries: Microscopic symmetries survive coarse-graining. In the case of an uniaxial magnet ($N = 1$) for example, in the absence of an external magnetic field, $F[\phi(\mathbf{r})]$ should have $\mathbb{Z}_2$ symmetry:

$$F[\phi(\mathbf{r})] = F[-\phi(\mathbf{r})],$$

For $N > 1$, $\mathbb{Z}_2$ is generalised to $O(N)$, that is, invariance under rotations and reflections:

$$F[\phi] = F[\mathbf{R}\phi]$$

with $\mathbf{R}^T\mathbf{R} = \mathbf{1}$. This implies that the Taylor expansion of $\phi(\mathbf{r})$ can only have terms even in $\phi(\mathbf{r})$: $\phi^2 = \phi \cdot \phi, \phi^4, \phi^6$ and so on.

As another example, if we are dealing with a symmetric binary fluid, this symmetry should be present too: the coarse-grained field can be taken as the difference between the fluid's densities. In the case of an asymmetric binary fluid such as a liquid-gas system, we no longer have $\mathbb{Z}_2$ symmetry and the free energy will acquire ϕ, ϕ^3 terms. However, if the fluid is conserved, we should have $\int d\mathbf{r}\phi =$ constant, so the linear term is trivial, and the cubic term can always be removed by a shift in ϕ and in the coefficient of ϕ^2, so that we recover the $\mathbb{Z}_2$ symmetry.

Moreover, the coarse-grained field can inherit the isotropy from microscopic models such as the uniaxial magnet:

$$F[\phi(\mathbf{R}\mathbf{r})] = F[\phi(\mathbf{r})]$$

where $\mathbf{R} \in O(d)$. So anisotropic terms such as $(\partial_1\phi)^2 + \kappa(\partial_2\phi)^2$ are forbidden. We can only have gradient terms like $(\nabla\phi)^2, (\nabla\phi)^4, (\nabla^2\phi)^2$ and so on, where

$$(\nabla\phi)^2 = \partial_i\phi_a\partial_i\phi_a$$

and the summation convention is implied.

Finally, in the presence of a magnetic field $h \neq 0$, $O(N)$ symmetry holds when a transformation $\phi \to R\phi$ is combined with $h \to Rh$, so the allowed terms in the free energy are $h \cdot \phi, h \cdot \phi\phi^2$ and so on.

(iv) Stability: $p[\phi(\mathbf{r})]$ should not diverge for unbounded $\phi(\mathbf{r})$. This means that $F[\phi(\mathbf{r})]$ should be positive and large for these configurations to be suppressed. Therefore the coefficient of the highest power of ϕ in the free energy should be positive.

The most general free energy that fulfils these criteria is the Landau-Ginzburg free energy:

$$\beta F[\phi(\mathbf{r})] = \int \left\{\frac{a}{2}\phi^2 + \frac{b}{4}\phi^4 + \frac{\kappa}{2}(\nabla\phi)^2 + ... - h \cdot \phi\right\} d\mathbf{r}. \tag{4}$$

The coefficients $a, b, \kappa, ...$ are functions of microscopic interactions and external parameters such as temperature and pressure. We can set $\beta = 1$ without loss of generality.

Depending on the behaviour of the coefficients $a, b, \kappa, ...$ and how many terms are included in the expansion, this free energy can describe a wide range of phenomena: from magnets and binary fluids to liquid crystals and superconductors.

2 Functional Methods

We wish to compute the partition function (3) with the Landau-Ginzburg free energy (4)

$$\beta F[\phi(\mathbf{r})] = \int \left\{f(\phi) + \frac{\kappa}{2}(\nabla\phi)^2\right\} d\mathbf{r}$$

where $f(\phi) = \frac{a}{2}\phi^2 + \frac{b}{4}\phi^4 - h \cdot \phi$. We further assume that $b, \kappa > 0$ and a is proportional to the system's reduced temperature $T - T_C$ for some critical temperature T_C.

In practice, (3) can only be evaluated exactly if the integrand is Gaussian. If higher order terms are small, it can be approximated using a perturbative approach.

2.1 Mean Field Theory

As an initial step we can perform a saddle point approximation, in which the integral is replaced by a sum over the set of extrema $\{\bar{\phi}\}$ of $\beta F[\phi(\mathbf{r})]$ (the saddle points):

$$Z \approx \sum_{\{\bar{\phi}\}} e^{-\beta F[\bar{\phi}]} \tag{5}$$

Note that $\bar{\phi}$ is the maximum of the probability distribution, so it corresponds to the equilibrium value of the field. It is known as the mean field. As $\int (\nabla\phi)^2 d\mathbf{r} \geq 0$, we can guess an uniform minimiser $\phi(\mathbf{r}) = \bar{\phi} \in \mathbb{R}^N$. The free energy (density) is then:

$$\frac{F}{V} = f(\bar{\phi}) = \frac{a}{2}\bar{\phi}^2 + \frac{b}{4}\bar{\phi}^4 - h \cdot \bar{\phi}$$

We can use this to analyse the behaviour of this system for small ϕ. As we would expect for a magnet, at high temperature ($a > 0$), if we drop the ϕ^4 term (because ϕ is small) there is a single minimum at

$$\bar{\phi} = \frac{1}{a}h \in \mathbb{R}^N.$$

As h is varied, $\bar{\phi}$ changes continuously and in the absence of an external field the mean field is just $\bar{\phi} = 0$. This is called the disordered phase.

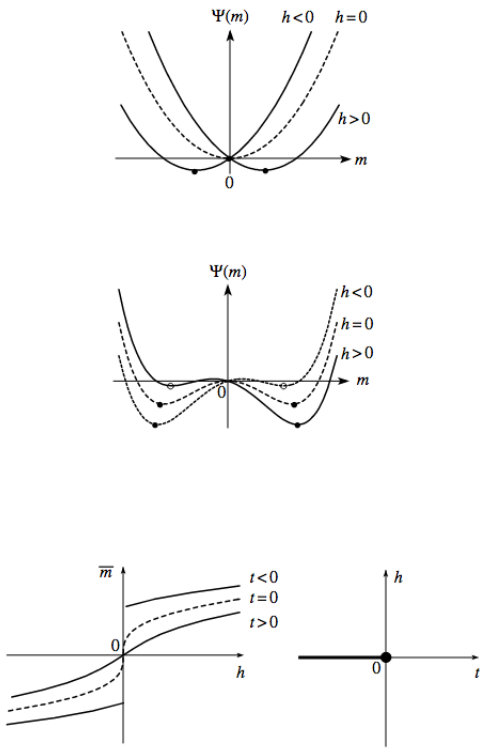


Figure 1: Free energy density, isotherms and phase diagram for the mean field theory of an uniaxial magnet (in this figure, $\phi = m, a = t$ and $f(\phi) = \Psi(m)$). At the top, we have $f(\phi)$ for $a > 0$, and at the middle $a < 0$. At the bottom left, we have $\bar{\phi}$ vs. h isotherms, displaying the discontinuous transition for $a < 0$ and $h \to 0$. The bottom right figure is the phase diagram for this system. Taken from [2].

However, for low temperatures ($a < 0$), the qualitative profile of the free energy changes drastically (see figure 1) and the ϕ^4 term becomes important to ensure stability. There are now two local minima, with the global one depending on the sign of h. We call this the ordered phase. For $h = 0$, the minima are degenerate:

$$\bar{\phi}^2 = -\frac{a}{b}.$$

Note that as $a \to 0$, $\bar{\phi}^2$ changes continuously from 0 to $-a/b$. This characterises a continuous phase transition. On the other hand, for $a < 0$, as $h \to 0$, $\bar{\phi}$ changes discontinuously. This is called a discontinuous phase transition.

These results are summarised in the isotherms and the phase diagram in figure 1. The line of discontinuous transitions terminates at the point $(a, h) = (0, 0)$, which is known as a critical point: at this point, the two phases are indistinguishable.

2.2 Spontaneous Symmetry Breaking

In the Landau-Ginzburg theory, when $a < 0, h = 0$, the minima are degenerate and the system has to "choose" one of them - we say that the symmetry is spontaneously broken. We have, up to a constant (see figure 2):

$$f(\phi) = \frac{b}{4}(\phi^2 - v^2)^2.$$

where $v^2 = \frac{-a}{b} > 0$. We can take $\bar{\phi} = (\theta_1, ..., \theta_{N-1}, v + \varphi) = (\theta, v + \varphi)$ as the mean field without loss of generality, where φ represents longitudinal and θ_i transverse modes.

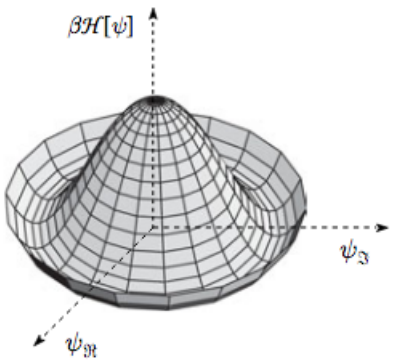


Figure 2: Landau-Ginzburg free energy of an O(2) field for $a < 0$. The symmetry is broken to $\{1\}$. The notation in this figure is $\mathcal{H}[\psi] = F[\phi]$ and $\psi = \phi$ in the main text.

Retaining only terms up to order $O(\varphi^2, \theta^2)$ we find

$$\begin{aligned} f(\phi) &= \frac{b}{4}\Big((v + \varphi)^2 + \theta^2 - v^2)^2 \\ &\approx f(0) + \frac{1}{2}\frac{\partial^2 f}{\partial \varphi^2}(0)\varphi^2 = bv^2\varphi^2 = -a\varphi^2 \end{aligned}$$

Therefore the longitudinal mode has quadratic coupling $-2a$ and we have found $N - 1$ "massless" transverse modes, i.e. terms with no quadratic coupling, which we call Goldstone modes.

In a more general context, phases of matter can be characterised by two symmetry groups. We let G be the symmetry of the free energy ($O(N)$ in the Landau-Ginzburg theory), and H the symmetry of the mean field. As an example, for a 3D magnet $G = O(3)$ and in the disordered phase $H = O(3)$ while in the ordered phase $H = O(2)$ (the mean field is invariant under a rotation along its parallel axis).

We expect now new configurations which minimise $f(\phi)$ but change direction in space. These configurations get energy from the gradient terms, but this can be lowered by increasing the wavelength of the fluctuations. The energy can then be arbitrarily small. These low energy excitations are the Goldstone modes and they arise in any system with a broken continuous symmetry.

Any two mean fields ϕ, ϕ' are related each other by a transformation in G, therefore for any $h \in H$

$$h\bar{\phi}' = \bar{\phi}' \Rightarrow hg\bar{\phi} = g\bar{\phi} \Rightarrow g^{-1}hg\bar{\phi} = \bar{\phi} \Rightarrow g^{-1}hg \in H$$

and we can conclude that the manifold of mean fields is isomorphic to the subgroup of G which leaves H invariant, $g^{-1}Hg = H$. This is equivalent to the coset space

$$G/H = O(N)/O(N-1) \cong S^{N-1}$$

where S^{N-1} is an $(N-1)$-sphere.

We get a Goldstone mode for each broken symmetry, so the number of Goldstone modes is $\dim G - \dim H = N - 1$. This is known as Goldstone's theorem. There are some exceptions to this theorem. For $N = 1$, it fails in one dimension, and for $N \geq 2$ it fails for $d \leq 2$. The latter is a consequence of the Mermin-Wagner theorem which we shall explore later in connection to fluctuations.

The cause of the failure of Goldstone's theorem in $d = 1$ for $N = 1$ is that we can find other free energy minima beyond the constant configurations $\bar{\phi}$. Using the Euler-Lagrange equation:

$$\begin{aligned}\frac{\delta F[\phi(\mathbf{r}]}{\delta\phi(\mathbf{r})} &= \frac{\partial f}{\partial \phi} - \kappa\nabla^2\phi \\ &= a\phi + b\phi^3 - \kappa\nabla^2\phi = 0\end{aligned}$$

We consider a function of the form $\phi(\mathbf{r}) = \phi(x)$, where x is one of the spatial directions, subject to $\phi(x) \xrightarrow{x\pm\infty} \pm\bar{\phi}$: this is a solution interpolating between $-\bar{\phi}$ and $\bar{\phi}$. This kind of solution connecting two minima is known in general as an instanton. We end up with:

$$a\phi + b\phi^3 - \kappa\frac{d^2\phi}{dx^2} = 0. \tag{6}$$

Solving this equation subject to the boundary conditions above results in a "domain wall" with the profile:

$$\phi(x) = \bar{\phi}\tanh\left(\frac{x - x_0}{W}\right)$$

where the width of the wall is $W = \sqrt{\frac{-2\kappa}{a}}$ and x_0 is an integration constant (figure 3).

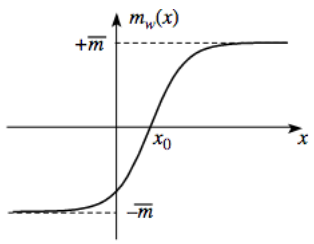


Figure 3: Example of an instanton solution ($m = \phi$ in the main text)

The excess free energy of the wall with respect to the mean field can be computed multiplying equation (6) by $d\phi/dx$ and integrating with respect to x. We find

$$\frac{\kappa}{2}\left(\frac{d\phi}{dx}\right)^2 = \frac{\kappa}{2}(\nabla\phi)^2 = a\frac{\phi^2}{2} + b\frac{\phi^4}{4}$$

$$\Rightarrow F_W = \kappa\int d\mathbf{r}(\nabla\phi)^2 = L^{d-1}\kappa\int dx\left(\frac{d\phi}{dx}\right)^2$$

where L is the size of the system. That is, F_W increases with the area of the domain wall L^{d-1}. Its probability is

$$p = \frac{e^{-\beta F_W}}{Z}\frac{L}{W}.$$

If we fix $\phi(-L/2) = \bar{\phi}$, the probability that $\phi(L/2) = \bar{\phi}$ equals the probability of an even number of walls between these two points. A wall can be found in any of the L/W possible positions so the probability of n walls scales with $L^n e^{-n\beta F_W}$

In particular, for $d = 1$, the fluctuations of domain walls are unsuppressed due to the entropy in their position - even for $a < 0$ the proliferation of domain walls washes away any order and there is no phase transition. However, for $d > 1$, the free energy scales with the domain wall's area, so domain walls are restrained and do not prevent a phase transition from occurring.

2.3 Critical Exponents

The singular behaviour of macroscopic quantities near a critical point can be characterised by a set of critical exponents denoted $\alpha, \beta, \gamma, \delta$ and defined by

$$\begin{aligned} c &\sim a^{-\alpha} \\ |\bar{\phi}| &\sim a^{\beta} \\ \chi &\sim a^{-\gamma} \\ |\bar{\phi}| &\sim h^{1/\delta} \end{aligned}$$

The exponents β and δ describe how the order parameter varies along phase transitions, and α, γ characterise the behaviour of response functions - functions that describe how the system responds to variations in the external parameters. The heat capacity c is the thermal response function and the susceptibility χ is the external field response function.

In the Landau-Ginzburg theory, the temperature dependence is

$$|\bar{\phi}| \sim \begin{cases} 0 & \text{for } a > 0, h = 0 \\ |a|^{\frac{1}{2}} & \text{for } a < 0, h = 0 \end{cases}$$

so we can take $\beta = \frac{1}{2}$. The magnetic field dependence is given by the minimum of $f(\bar{\phi}, a = 0)$:

$$f'(\bar{\phi}, a = 0) = b|\bar{\phi}|^3 - h = 0 \Rightarrow |\bar{\phi}| \sim h^{\frac{1}{3}}$$

therefore $\delta = 3$. The heat capacity is

$$c(h = 0) \equiv \frac{1}{N}\frac{\partial E}{\partial T} = \frac{\beta^2}{N}\frac{\partial^2 logZ}{\partial \beta^2} = -T\frac{\partial^2 F}{\partial T^2}$$

using that the free energy is

$$F(h = 0) = Vf(\bar{\phi}, h = 0) = \begin{cases} 0 & \text{for } a > 0 \\ \frac{-a^2}{4b} & \text{for } a < 0 \end{cases}$$

$$\Rightarrow c \sim \begin{cases} 0 & \text{for } a > 0 \\ \frac{1}{b} & \text{for } a < 0 \end{cases}$$

that is, c jumps discontinuously. We can say that $\alpha = 0$(disc.) to emphasise this. Lastly, the magnetic susceptibility is

$$\chi \equiv \frac{\partial |\bar{\phi}|}{\partial h}\Big|_{h=0}$$

and it can be obtained from $|\bar{\phi}| = |h|/a$ for $a > 0$, For $a < 0$, we need to restore the quartic term and expand the free energy near $\bar{\phi}$ to second order (so that we can find f' to first order and find the

minimum):

$$
\begin{aligned}
f(\bar{\phi}+\delta\phi) &= f(\bar{\phi}) + f'(\bar{\phi})\delta\phi + \frac{1}{2}f''(\bar{\phi})\delta\phi^2 \\
\Rightarrow f'(\bar{\phi}+\delta\phi) &= f'(\bar{\phi}) + f''(\bar{\phi})\delta\phi = 0 \\
\Rightarrow \cancel{a|\bar{\phi}|} + \cancel{b|\bar{\phi}|^3} - h + a\delta\phi + 3b|\bar{\phi}|^2\delta\phi &= 0
\end{aligned}
$$

and therefore $\delta\phi = -\frac{h}{2a}$. We find

$$\chi \sim |a|^{-1}$$

therefore $\gamma = 1$. To summarise, the mean field results are displayed in the table below, together with the experimental results for dimensions $d = 2, 3$.

	$d \geq 4$ (MF)	$d = 2$	$d = 3$
α	0 (disc.)	0 (log.)	0.1101...
β	1/2	1/8	0.3264...
γ	1	7/4	1.2371...
δ	3	15	4.7898...

Table 1: Mean Field Theory and experimental values for the critical exponents of the Ising model ($N = 1$).

Note that the mean field results are correct for $d \geq 4$ (this is called the upper critical dimension). They fail completely for $d = 1$ (the lower critical dimension) as expected from our discussion of last section. For $d = 2, 3$, although the critical exponents are wrong, the phase diagram is correct. To correctly analyse these cases we need to take fluctuations into account.

Experimental results suggest that these exponents are universal: they independ on the material and nature of the phase transition. This is known as universality. For example, consider the liquid-gas phase diagram. This is quite similar to the uniaxial magnet diagram - a line of 1st order phase transitions ending in a critical point (figure 4).

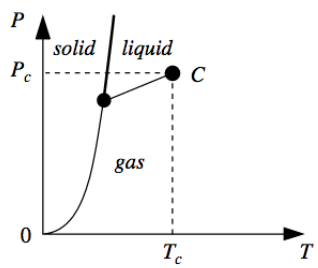


Figure 4: Phase diagram for the liquid-gas transition. P, T denotes pressure and temperature, and C is the critical point. Taken from [2].

The uniaxial magnet diagram in figure 1 appears simpler because the $\mathbb{Z}_2$ symmetry ensures that the critical point occurs at $h = 0$. If we consider the fractional density of liquid and gas as our order parameter, the pressure as the external field (analogous to h in the magnet) and use the Van der Waals equation to compute the heat capacity and compressibility (analogous to the susceptibility), we find the exact same critical exponents.

2.4 The Gaussian Model

Now we move on to consider cases where the partition function can be computed exactly - Gaussian models. These have free energies of the form

$$\beta F[\phi(\mathbf{r})] = \frac{1}{2}\int d\mathbf{r}\{a\phi(\mathbf{r})^2 + \kappa(\nabla\phi)^2\}$$

with all terms quadratic in the field.

It will be convenient to have the Fourier transform of the free energy. For this we first Fourier transform the field:

$$\phi_{\mathbf{k}} = \int d\mathbf{r} e^{-i\mathbf{k}\cdot\mathbf{r}} \phi(\mathbf{r}),$$
$$\phi(\mathbf{r}) = \int \frac{d\mathbf{k}}{(2\pi)^d} e^{i\mathbf{k}\cdot\mathbf{r}} \phi_{\mathbf{k}}.$$

Note that as $\phi(\mathbf{r}) \in \mathbb{R}$, $\phi^*_{\mathbf{k}} = \phi_{-\mathbf{k}}$. Now we can write the free energy as

$$\beta F[\phi_{\mathbf{k}}] = \frac{1}{2} \int \frac{d\mathbf{k}_1}{(2\pi)^d} \frac{d\mathbf{k}_2}{(2\pi)^d} d\mathbf{r} (a - \kappa \mathbf{k}_1 \cdot \mathbf{k}_2) \phi_{\mathbf{k}_1} \cdot \phi_{\mathbf{k}_2} e^{i(\mathbf{k}_1+\mathbf{k}_2)\cdot\mathbf{r}}$$

and finally, using that

$$\int d\mathbf{r} e^{i\mathbf{k}\cdot\mathbf{r}} = (2\pi)^d \delta^{(d)}(\mathbf{k})$$

we find

$$\beta F[\phi_{\mathbf{k}}] = \frac{1}{2} \int \frac{d\mathbf{k}}{(2\pi)^d} (a + \kappa \mathbf{k}^2) |\phi_{\mathbf{k}}|^2.$$

If we put this back into the partition function, we see that it becomes simply a Gaussian integral. We consider a system of finite volume V, such that

$$\beta F[\phi_{\mathbf{k}}] = \frac{1}{2V} \sum_{\mathbf{k}} (a + \kappa \mathbf{k}^2) |\phi_{\mathbf{k}}|^2$$

then the partition function decouples to a Gaussian integral for each component of the field:

$$\begin{aligned} Z &= \int \Big(\prod_{\mathbf{k}} d\phi_{\mathbf{k}} \Big) \exp\Big(-\frac{\beta}{2V} \sum_{\mathbf{k}} (a + \kappa \mathbf{k}^2) |\phi_{\mathbf{k}}|^2 \Big) \\ &= \prod_{i,k} \int d\phi^i_{\mathbf{k}} \exp\Big(-\frac{\beta}{2V} (a + \kappa \mathbf{k}^2) |\phi^i_{\mathbf{k}}|^2 \Big) \\ &= \left(\prod_{\mathbf{k}} \sqrt{\frac{2\pi V}{\beta(a + \kappa \mathbf{k}^2)}} \right)^N \end{aligned}$$

with the caveat that, in the $a < 0$ we would have to distinguish longitudinal and transverse modes for which the value of a changes. We see that each mode $\phi_{\mathbf{k}}$ behaves as a random Gaussian variable with mean value 0, therefore:

$$\langle |\phi(\mathbf{r})| \rangle = \begin{cases} 0 \text{ if } a > 0 \\ \bar{\phi} \text{ if } a < 0 \end{cases}$$

From this we can compute, for example, the α exponent for the Gaussian model. We start with the Helmholtz free energy density:

$$\frac{F_H}{V} = -\frac{T}{V} \log Z = -\frac{NT}{2V} \sum_{\mathbf{k}} \log \frac{2\pi V T}{(a + \kappa \mathbf{k}^2)}$$

or in the continuum limit (up to a constant):

$$\frac{\log Z}{V} = -\frac{N}{2} \int \frac{d\mathbf{k}}{(2\pi)^d} \log(a + \kappa \mathbf{k}^2) = -\frac{N}{2} a^{d/2} \int \frac{d\mathbf{k}'}{(2\pi)^d} \log a(1 + \kappa \mathbf{k}'^2)$$

where I have set $\mathbf{k}' = \mathbf{k}/\sqrt{a}$. Now we can compute how the heat capacity scales with temperature:

$$c \sim \frac{\partial^2 F}{\partial a^2} \sim -\frac{N}{2} a^{\frac{d}{2}-2} \int \frac{d\mathbf{k}}{(2\pi)^d} \frac{1}{(1+\kappa \mathbf{k}^2)^2}$$

where I used that $E = -\partial_\beta \log Z$. The behaviour for $a < 0$ is qualitatively the same but the contribution of the Goldstone modes does not depend on a so we do not get the factor of N.

For $d > 4$, the integral diverges so the leading contribution to c is independent of a. But for $d < 4$, the leading contribution is

$$c \sim a^{\frac{d-4}{2}}$$

therefore we find $\alpha = 2 - \frac{d}{2}$. Near the critical point, the critical exponent will receive further corrections from higher order terms that are necessary to make the free energy stable.

So there is a qualitative change to the critical exponent at $d = 4$, from a discontinuity ($\alpha = 0$) to a divergence ($\alpha > 1$). The fluctuations are responsible for the inconsistencies between MF results and experimental observations below the upper critical dimension $d = 4$ (table 1).

2.5 Correlation Functions

Fluctuations about the mean field are captured by the 2-point correlation function, $\langle \phi^a(\mathbf{r}_1)\phi^b(\mathbf{r}_2)\rangle$. To compute this, the starting point is the free energy with a varying external field:

$$\beta F[\phi(\mathbf{r})] = \int \Big\{ \frac{a}{2}\phi(\mathbf{r})^2 + \frac{\kappa}{2}(\nabla\phi)^2 - h(\mathbf{r}) \cdot \phi(\mathbf{r}) \Big\} d\mathbf{r}$$

with corresponding partition function

$$Z[h] = \int D\phi e^{-\beta F[\phi]}.$$

The correlation function can be found by taking derivatives of $Z[h]$ with respect to h:

$$\langle \phi^a(\mathbf{r}_1)\phi^b(\mathbf{r}_2)\rangle = \frac{1}{\beta^2} \frac{\delta^2}{\delta h^a(\mathbf{r}_1)\delta h^b(\mathbf{r}_2)} \frac{Z[h]}{Z}\bigg|_{h=0} = \frac{1}{Z}\int D\phi\ \phi^a(\mathbf{r}_1)\phi^b(\mathbf{r}_2) e^{-\beta F[\phi]}.$$

So we have to evaluate $Z[h]$. This is more easily done in Fourier space, where the free energy is

$$\beta F[\phi_\mathbf{k}] = \int \frac{d\mathbf{k}}{(2\pi)^d} \Big\{ \frac{1}{2}(a + \kappa\mathbf{k}^2)|\phi_\mathbf{k}|^2 - h_\mathbf{k} \cdot \phi_{-\mathbf{k}} \Big\}$$

or, changing variables $\phi_\mathbf{k} \to \phi_\mathbf{k} + \frac{h_\mathbf{k}}{a+\kappa\mathbf{k}^2}$

$$\beta F[\phi_\mathbf{k}] = \int \frac{d\mathbf{k}}{(2\pi)^d} \Big\{ \frac{1}{2}(a + \kappa\mathbf{k}^2)|\phi_\mathbf{k}|^2 - \frac{|h_\mathbf{k}|^2}{a + \kappa\mathbf{k}^2} \Big\}$$

such that $Z[h]$ is Gaussian:

$$Z[h] = \int \Big(\prod_\mathbf{k} d\phi_\mathbf{k} \Big) e^{-\beta F[\phi_\mathbf{k}]} = e^{-\beta F_H} \exp\Big(\frac{\beta}{2} \int \frac{d\mathbf{k}}{(2\pi)^d} \frac{|h_\mathbf{k}|^2}{a + \kappa\mathbf{k}^2} \Big)$$

where F_H is the Helmholtz free energy for the Gaussian model, evaluated in the last section. Transforming back to real space, we find

$$Z[h] = e^{-\beta F_H} \exp\Big(\frac{\beta}{2} \int d\mathbf{r}_1 d\mathbf{r}_2 h^a(\mathbf{r}_1) G_{ab}(\mathbf{r}_1, \mathbf{r}_2) h^b(\mathbf{r}_2) \Big)$$

where the Green function is defined as

$$G_{ab}(\mathbf{r}_1, \mathbf{r}_2) = G_{ab}(\mathbf{r}_1 - \mathbf{r}_2) \equiv \int \frac{d\mathbf{k}}{(2\pi)^d} \frac{\delta_{ab} e^{-i\mathbf{k}\cdot(\mathbf{r}_1 - \mathbf{r}_2)}}{a + \kappa \mathbf{k}^2}.$$

The correlation function is then

$$\langle \phi^a(\mathbf{r}_2)\phi^b(\mathbf{r}_2)\rangle = \frac{1}{\beta} G_{ab}(\mathbf{r}_1 - \mathbf{r}_2) = \frac{1}{\beta\kappa} \int \frac{d\mathbf{k}}{(2\pi)^d} \frac{\delta_{ab} e^{-i\mathbf{k}\cdot(\mathbf{r}_1 - \mathbf{r}_2)}}{\xi^{-2} + \mathbf{k}^2}$$

where we have defined the correlation length $\xi^2 = \kappa/a$. It can be shown that ([2])

$$\langle \phi^a(\mathbf{r}_1)\phi^b(\mathbf{r}_2)\rangle \sim \begin{cases} \delta_{ab} r^{2-d} \text{ for } r \ll \xi \\ \delta_{ab} e^{-r/\xi}/r^{\frac{d-1}{2}} \text{ for } r \gg \xi \end{cases}.$$

This means that the field is significantly correlated over length scales of ξ or less (thus the name correlation length). Note that $\xi^2 \sim a^{-1}$, so at the critical point $\xi \to \infty$, meaning that there are fluctuations at all length scales. Of course, at $a = 0$ we have to consider higher order terms to make the theory stable again.

Finally, we note that Goldstone modes have infinite correlation length: their 2-point function is given by

$$\langle \theta^i(\mathbf{r}_1)\theta^i(\mathbf{r}_2)\rangle = \frac{1}{\beta\kappa} \int \frac{d\mathbf{k}}{(2\pi)^d} \frac{e^{-i\mathbf{k}\cdot(\mathbf{r}_1 - \mathbf{r}_2)}}{\mathbf{k}^2}$$

this is similar to the correlation function at the critical point. This has the important consequence that, for fields with $N > 1$, the lower critical dimension is 2 instead of 1. Indeed, for $d \leq 2$ the integral above grows with $\mathbf{r}_1 - \mathbf{r}_2$, destroying the ordered phase - there is no symmetry breaking of continuous symmetries in $d = 1, 2$, and thus there are no Goldstone modes. This general result is known as the Mermin-Wagner theorem.

3 Renormalization Group

The Landau-Ginzburg theory (4) is parametrised by an infinite number of coupling constants $\{g_n\}$, which we can regard as coordinates on an infinite dimensional "theory space" and a UV cutoff Λ, that is, a wavevector above which the field cannot vary:

$$\phi_{\mathbf{k}} = 0 \text{ for } |\mathbf{k}| > \Lambda.$$

We can think of universality as the fact that at long distance (IR) scales the theory should make the same predictions independent of the degrees of freedom that vary on scales close to Λ (UV). The renormalization group is the method that allows us to study how theories with different short-distance details can give rise to the same long-distance physics.

We consider again the full Landau-Ginzburg free energy:

$$\beta F_{\Lambda_0}[\phi(\mathbf{r})] = \int \Big\{ \frac{1}{2}(\nabla\phi)^2 + \frac{a}{2}\phi^2 + \sum_{n \text{ even}>2} g_{n,0}\ \phi^n \Big\} d\mathbf{r},$$

where we have set $\kappa = 1$ (this is called a canonical kinetic term) and we assume all temperature dependence to be in a. The partition function with a cutoff Λ_0 is

$$Z_{\Lambda_0}(g_{n,0}) = \int^{\Lambda_0} D\phi e^{-\beta F_{\Lambda_0}[\phi]},$$

where we denote $g_2 = a/2$. The integral is taken over field modes with $|\mathbf{k}| < \Lambda_0$, that is

$$\phi(\mathbf{r}) = \int_{|\mathbf{k}|\leq\Lambda_0} \frac{d\mathbf{k}}{(2\pi)^d} e^{i\mathbf{k}\cdot\mathbf{r}} \phi_{\mathbf{k}}.$$

We can split these modes into IR and UV parts selecting a new scale $\Lambda < \Lambda_0$

$$\phi(\mathbf{r}) = \int_{|\mathbf{k}|\leq\Lambda} \frac{d\mathbf{k}}{(2\pi)^d} e^{i\mathbf{k}\cdot\mathbf{r}} \phi_{\mathbf{k}} + \int_{\Lambda\leq|\mathbf{k}|\leq\Lambda_0} \frac{d\mathbf{k}}{(2\pi)^d} e^{i\mathbf{k}\cdot\mathbf{r}} \phi_{\mathbf{k}} \equiv \phi^-(\mathbf{r}) + \phi^+(\mathbf{r}).$$

The measure factorises because the two regions are disjoint, $D\phi = D\phi^- D\phi^+$. If we integrate out the UV modes, we obtain an effective free energy with a lower UV cutoff Λ. This corresponds to a coarse-graining procedure, eliminating fluctuations below a length scale of Λ^{-1}:

$$e^{-\beta F_\Lambda[\phi^-]} = \int_{\Lambda\leq|\mathbf{k}|\leq\Lambda_0} D\phi^+ e^{-\beta F_{\Lambda_0}[\phi^- + \phi^+]},$$

such that

$$Z_\Lambda(g_n(\Lambda)) = \int^\Lambda D\phi e^{-\beta F_\Lambda[\phi]}.$$

As F_{Λ_0} already had the most general form possible and coarse-graining retains the original symmetries, F_Λ must take the form

$$F_\Lambda[\phi] = \int \Big\{ \frac{\kappa'}{2}(\nabla\phi)^2 + \frac{a'}{2}\phi^2 + \sum_{n \text{ even}>2} g'_n \, \phi^n \Big\} d\mathbf{r}.$$

We cannot immediately compare the new coefficients with the old ones because the theories are defined at different cutoffs. First we define $\zeta = \frac{\Lambda_0}{\Lambda} > 1$ and rescale the momenta

$$\mathbf{k}' = \zeta\mathbf{k}.$$

This has the effect of scaling the cutoff back to Λ. In real space, this is equivalent to rescaling spatial coordinates

$$e^{i\mathbf{k}\cdot\mathbf{r}} = e^{i\mathbf{k}'\cdot\mathbf{r}'} \iff \mathbf{r}' = \frac{\mathbf{r}}{\zeta}$$

that is, we are "zooming out" to see the theory at a large scale. We can think of this as if coarse-graining had reduced the "resolution" of the field, and rescaling restores it by shrinking all length scales.

The last step is to rescale the field itself, in a way to fix $\kappa = 1$:

$$\phi' = \sqrt{\kappa'}\phi$$

we can interpret this as restoring the "contrast" that coarse-graining decreased. The resulting free energy is (dropping the primes):

$$F_\Lambda[\phi] = \int \Big\{ \frac{1}{2}(\nabla\phi)^2 + \frac{a(\zeta)}{2}\phi^2 + \sum_{n \text{ even}>2} g_n(\zeta) \, \phi^n \Big\} d\mathbf{r}.$$

These 3 steps give us what is known as the renormalization flow in theory space - given a set of parameters $(a, g_4, ...)$, it maps it to a new set $(a(\zeta), g_4(\zeta), ...)$. Summarising, the 3 steps of RG are:

(i) Integrate out UV modes (coarse-grain)

(ii) Rescale $\mathbf{k}' = \zeta\mathbf{k}$ (zoom out)

(iii) Renormalize fields to get $\kappa = 1$ (restore contrast)

Note that the RG flow is not really a group: step 1 is not invertible because we lose some information with coarse-graining.

A fixed point (or critical point) is defined as a point in theory space, (a^*, g_n^*) that is mapped onto itself by the renormalization flow. Since all length scales are shrunk under RG, the correlation length transforms as

$$\xi \to \xi' = \frac{\xi}{\zeta}$$

therefore at a fixed point we must have $\xi' = \xi$. The only possibilities for that are either $\xi = 0$ or $\xi = \infty$. As $\xi^2 \sim a^{-1}$, the former happens in the limit of infinite temperature $a \to \infty$ (complete disorder). More interestingly, the fixed point at $\xi = \infty$ corresponds to the critical point $a = 0$, where there are fluctuations on all length scales.

We can classify parameters (couplings) depending on their flows near fixed points.

(i) Couplings that flow back to the fixed point under RG are irrelevant. Turning on these interactions doesn't change the long distance physics.

(ii) Couplings that flow away from the fixed point when turned on are relevant.

(iii) If our fixed points form a line or surface in theory space, moving couplings along this surface barely changes IR physics. These directions are called marginal.

In general, the fixed point has a finite number of relevant directions and infinite irrelevant ones. This is the reason for universality - including all of the microscopic details in a model puts it in a particular point in theory space. But as we flow towards the IR, many theories converge to the same fixed point, with the same critical exponents.

3.1 Scaling

Some of the ideas underlying the RG can be used to build relations between the critical exponents seen in section 2.3. At the fixed point, there is no length scale, so the correlation function must take the form of a power law (any other form would require a scale on dimensional grounds):

$$G(r) \sim \frac{1}{r^{d-2+\eta}}.$$

This is similar to the power law found in section 2.5 for $r \ll \xi$, but experiment tells us that there is an extra anomalous dimension η when $d = 2, 3$. By naive dimensional analysis, looking at the kinetic term, we would obtain

$$[F] = L^0 \Rightarrow [\phi] = L^{\frac{2-d}{2}} \Rightarrow \eta = 0$$

because $G(r)$ has two factors of ϕ.

However, the naive analysis fails when we have an UV cutoff in our theory - this introduces a further scale in the problem. As in the 3rd step of RG the field is rescaled in a way that depends on Λ, we cannot perform naive dimensional analysis. We define a new "scaling dimension", which tells us how the quantities change under scaling:

$$\begin{aligned}
&\mathbf{r} \to \zeta^{\Delta_{\mathbf{r}}}\mathbf{r} \Rightarrow \Delta_{\mathbf{r}} = -1 \\
&\mathbf{k} \to \zeta^{\Delta_{\mathbf{k}}}\mathbf{k} \Rightarrow \Delta_{\mathbf{k}} = 1 \\
&\phi \to \zeta^{\Delta_\phi}\phi.
\end{aligned}$$

The scaling dimension Δ_ϕ can be obtained from the expression above for $G(r) = \langle\phi(0)\phi(\mathbf{r})\rangle$,

$$\Delta_\phi = \frac{d-2+\eta}{2}.$$

Besides η, we define the exponent ν as

$$\xi \sim a^{-\nu}.$$

In section 2.5 we defined $\xi^2 \sim a^{-1}$ so in the Gaussian approximation $\nu = -\frac{1}{2}$. This can be used to find the scaling dimension of a:

$$\xi \to \frac{\xi}{\zeta} \Rightarrow \Delta_\xi = -1,$$
$$a \to \zeta^{\Delta_a} a \Rightarrow \Delta_a = \frac{1}{\nu}.$$

It turns out that scaling arguments such as the one above allow us to establish relations giving all critical exponents in terms of only two of them. We choose these two to be η and ν (see table 2).

	$d \geq 4$ (MF)	$d = 3$	$d = 2$
ν	1/2	0.6300	1
η	0	0.0363	1/4

Table 2: MF and experimental results for ν, η.

To find α we ask how the free energy scales with a. As ξ is the only length scale and F is an extensive, dimensionless quantity, it must be proportional to $(L/\xi)^d$, where L is the size of the system. So we find $F \sim a^{d\nu}$ and therefore

$$c \sim \frac{\partial^2 F}{\partial a^2} \sim t^{d\nu - 2}.$$

Similarly,

$$\phi \sim a^\beta \Rightarrow \Delta_\phi = \frac{\beta}{\nu}$$
$$\chi = \frac{\partial \phi}{\partial h}\Big|_{a=0} \sim a^{-\gamma} \Rightarrow \Delta_\phi - \Delta_h = -\gamma \Delta_a$$
$$\phi \sim h^{\frac{1}{\delta}} \Rightarrow \delta = \frac{\Delta_h}{\Delta_\phi}$$

and the scaling dimension of h is

$$F \sim \int d\mathbf{r} h\phi \Rightarrow \Delta_\phi + \Delta_h = d \Rightarrow \Delta_h = \frac{d + 2 - \eta}{2}$$

Thus we find the relations

$$\alpha = 2 - d\nu$$
$$\beta = \frac{(d - 2 + \eta)\nu}{2}$$
$$\gamma = \nu(2 - \eta)$$
$$\delta = \frac{d + 2 - \eta}{d - 2 + \eta}$$

The scaling dimension can be used to characterise interactions, for instance

$$\int d\mathbf{r} g_n \phi^n.$$

Under scaling $\mathbf{r} \to \mathbf{r}/\zeta$ we have

$$\phi^n \to \zeta^{n\Delta_\phi} \phi^n$$

therefore the coupling will scale as $g_n \to \zeta^{d - n\Delta_\phi} g_n$ to keep the free energy dimensionless. Thus we can infer that an interaction is:

(i) relevant if $n\Delta_\phi < d$ (i.e. g_n grows under RG)

(ii) irrelevant if $n\Delta_\phi > d$ (i.e. g_n shrinks under RG)

(iii) marginal if $n\Delta_\phi = d$.

3.2 RG Analysis of the Gaussian Model

We consider again the Gaussian model (section 2.4) with the canonical kinetic term

$$\beta F_{\Lambda_0}[\phi] = \frac{1}{2}\int \left\{a\phi^2 + (\nabla\phi)^2\right\} d\mathbf{r}$$

and start by transforming to Fourier space, obtaining

$$\beta F_{\Lambda_0}[\phi_{\mathbf{k}}] = \frac{1}{2}\int_{\Lambda_0} \frac{d\mathbf{k}}{(2\pi)^d}(a + \mathbf{k}^2)|\phi_{\mathbf{k}}|^2.$$

Now we repeat the 3 steps of RG, starting by integrating out UV modes. Note that the free energy above decouples because the UV and IR modes have different support in Fourier space:

$$F_{\Lambda_0}[\phi^+ + \phi^-] = F_{[\Lambda,\Lambda_0]}[\phi^+] + F_\Lambda[\phi^-]$$

where the subscripts of the F's denote the support of integral. Therefore the effective free energy is

$$e^{-\beta F_\Lambda[\phi]} = \Big(\int D\phi^+ e^{-\beta F_{[\Lambda,\Lambda_0]}[\phi^+]}\Big) e^{-\beta F_\Lambda[\phi]} = N e^{-\beta F_\Lambda[\phi]}.$$

where N is a constant. The 2nd step consists in rescaling and the 3rd renormalizing:

$$\mathbf{k}' = \zeta\mathbf{k}$$
$$\phi'_{\mathbf{k}'} = \zeta^{\Delta_\phi - d}\phi_{\mathbf{k}}$$

where the $\Delta_\phi - d$ scaling comes from taking the Fourier transform. We end up with

$$\beta F_\Lambda[\phi'_{\mathbf{k}'}] = \frac{1}{2}\int_\Lambda \frac{d\mathbf{k}'}{(2\pi)^d}\zeta^{-d}(a + \zeta^{-2}\mathbf{k}'^2)|\phi'_{\mathbf{k}}|^2\zeta^{2(d-\Delta_\phi)}.$$

Requiring a canonical kinetic term, we find

$$\Delta_\phi = \frac{d-2}{2} \Rightarrow \eta = 0$$

so the anomalous dimension is 0 for the Gaussian theory. Finally, we can rewrite the free energy in terms of the rescaled couplings (dropping the primes)

$$\beta F_\Lambda[\phi_{\mathbf{k}}] = \frac{1}{2}\int_\Lambda \frac{d\mathbf{k}}{(2\pi)^d}(a(\zeta) + \mathbf{k}^2)|\phi_{\mathbf{k}}|^2$$

where

$$a(\zeta) = \zeta^2 a \qquad \text{(relevant)}.$$

Note that as $\xi^2 \sim a^{-1}$ we find $\xi \to \xi/\zeta$ as expected. The fixed point, known as Gaussian fixed point (GFP) is $a = 0$ and the critical exponents are $\nu = \frac{1}{2}$, $\eta = 0$ which agree with the previous section.

Now let us consider what happens if we add interactions to the free energy

$$\beta F[\phi] \to \beta F[\phi] + \int d\mathbf{r} \sum_{n \text{ even}>2} g_{n,0}\ \phi^n.$$

Now the 1st step of RG is not so simple because the Fourier transform of ϕ^n mixes IR and UV modes (we get terms like $\phi^+_{\mathbf{k}_1}...\phi^-_{\mathbf{k}_i}$). Integrating out the UV modes, the coefficients will change as

$$g_{n,0} \to g'_{n,0} = g_{n,0} + \delta g_n.$$

We will learn how to compute δg_n in the next section. We shall ignore it for now and focus on the 2nd and 3rd steps of RG (rescale and renormalize) which bring the interaction to the form

$$\int d\mathbf{r}\zeta^d \sum_{n \text{ even}>2} \zeta^{-n\Delta_\phi} g_{n,0}\ \phi^n = \int d\mathbf{r} \sum_{n \text{ even}>2} g_n(\zeta)\ \phi^n$$

where

$$g_n(\zeta) = \zeta^{n+d-nd/2} g_{n,0}$$

Note that, for $n = 4$, $g_4(\zeta) = \zeta^{4-d} g_{4,0}$. Therefore the ϕ^4 coupling is relevant for $d < 4$. It seems to be marginal for $d = 4$, but we will see in the next section that we need to take δg_4 into account. All higher order couplings are irrelevant for $d = 4$.

This explains why mean field theory fails for $d < 4$. If we are in $d > 4$, we can reach the GFP by tuning one parameter (a) because ϕ^4 is irrelevant. But for low dimensions, the GFP is unstable because of the relevance of the quartic coupling - it is hard to reach it experimentally because the g_4 IR flow pushes us away. To see exactly where the theory flows to in $d < 4$, we need to work perturbatively, which is the subject of the next section.

4 Perturbative Renormalization Group

Let us denote the Gaussian free energy by $F_0[\phi]$. We want to consider the Landau-Ginzburg theory as a perturbation to the Gaussian model of the form

$$F[\phi] = F_0[\phi] + F_I[\phi]$$

where, as we expect all higher order couplings to be irrelevant, we take

$$F_I[\phi] = \int d\mathbf{r} g\phi^4 = g\int d\mathbf{r}(\phi\cdot\phi)^2 = g\int d\mathbf{r}\delta_{ab}\delta_{cd}\phi^a\phi^b\phi^c\phi^d.$$

It is in fact more useful to work in Fourier space where

$$\begin{aligned} F_I[\phi] &= g\delta_{a_1a_2}\delta_{a_3a_4}\int d\mathbf{r}\int \frac{d\mathbf{k}_1\ d\mathbf{k}_2\ d\mathbf{k}_3\ d\mathbf{k}_4}{(2\pi)^{4d}} e^{-i\mathbf{r}\cdot(\mathbf{k}_1+\mathbf{k}_2+\mathbf{k}_3+\mathbf{k}_4)}\phi^{a_1}_{\mathbf{k}_1}\phi^{a_2}_{\mathbf{k}_2}\phi^{a_3}_{\mathbf{k}_3}\phi^{a_4}_{\mathbf{k}_4} \\ &= g\delta_{a_1a_2}\delta_{a_3a_4}\int \frac{d\mathbf{k}_1\ d\mathbf{k}_2\ d\mathbf{k}_3}{(2\pi)^{3d}}\phi^{a_1}_{\mathbf{k}_1}\phi^{a_2}_{\mathbf{k}_2}\phi^{a_3}_{\mathbf{k}_3}\phi^{a_4}_{-\mathbf{k}_1-\mathbf{k}_2-\mathbf{k}_3} \end{aligned} \tag{7}$$

To analyse the RG flow of this theory, we will have to compute various expectation values. In the next section we develop some techniques to do this.

4.1 Expectation Values

First, we show how to compute expectation values in the unperturbed Gaussian model. We write

$$\beta F_0[\phi_{\mathbf{k}}] = \frac{1}{2}\int \frac{d\mathbf{k}}{(2\pi)^d}(a+\mathbf{k}^2)|\phi_{\mathbf{k}}|^2.$$

The Fourier space 2-point correlation function can be obtained by Fourier transforming $G_{ab}(\mathbf{r}_1,\mathbf{r}_2)$ (section 2.5):

$$\langle\phi^a_{\mathbf{k}}\phi^b_{\mathbf{k}'}\rangle_0 = \frac{(2\pi)^d\delta^{(d)}(\mathbf{k}+\mathbf{k}')\delta_{ab}}{a+\mathbf{k}^2}$$

where the subscript denotes an expectation value with respect to the Gaussian free energy.

To compute correlation functions of higher order, we will make use of a result known as Wick's theorem. It can be derived from the identity

$$\langle e^{h_a\phi_a}\rangle_0 = \frac{Z[h]}{Z} = e^{\frac{1}{2}h_a\langle\phi_a\phi_b\rangle_0 h_b}$$

where we assume the summation convention and we discretised all of the integrals for readability. $Z[h]$ was defined in section 2.5. Expanding both sides in powers of h_a, we find

$$\langle e^{h_a\phi_a}\rangle_0 = 1 + h_a\langle\phi_a\rangle_0 + \frac{1}{2}h_a h_b\langle\phi_a\phi_b\rangle + \frac{1}{3!}h_a h_b h_c\langle\phi_a\phi_b\phi_c\rangle_0 + \frac{1}{4!}h_a h_b h_c h_d\langle\phi_a\phi_b\phi_c\phi_d\rangle_0 + ...$$

and

$$e^{\frac{1}{2}h_a\langle\phi_a\phi_b\rangle_0 h_b} = 1 + \frac{1}{2}h_a h_b\langle\phi_a\phi_b\rangle_0 + \frac{1}{4!}h_a h_b h_c h_d\big(\langle\phi_a\phi_b\rangle_0\langle\phi_c\phi_d\rangle_0 + \langle\phi_a\phi_c\rangle_0\langle\phi_b\phi_d\rangle_0 + \langle\phi_a\phi_d\rangle_0\langle\phi_b\phi_c\rangle_0\big) + ...$$

where we have used that

$$\begin{aligned}
\partial^4_{abcd}\Big(e^{\frac{1}{2}h_e M_{ef} h_f}\Big)_{h=0} &= \partial^3_{abc}\Big(M_{dg}h_g e^{\frac{1}{2}h_e M_{ef}h_f}\Big)_{h=0} = \partial^2_{ab}\Big\{\big(M_{dc} + M_{dg}h_g M_{ci}h_i\big)e^{\frac{1}{2}h_e M_{ef}h_f}\Big\}_{h=0} \\
&= \partial_a\Big\{\big(M_{dc}M_{bg}h_g + M_{db}M_{ci}h_i + M_{dg}h_g M_{cb} + M_{dg}h_g M_{ci}h_i M_{bj}h_j\big)e^{\frac{1}{2}h_e M_{ef}h_f}\Big\}_{h=0} \\
&= M_{ab}M_{cd} + M_{ac}M_{bd} + M_{ad}M_{bc}
\end{aligned}$$

so comparing both sides order by order in h, we can identify (back to the standard notation)

$$\begin{aligned}
\langle\phi_{\mathbf{k}}\rangle_0 &= 0 \\
\langle\phi_{\mathbf{k}_1}\phi_{\mathbf{k}_2}\phi_{\mathbf{k}_3}\rangle_0 &= 0 \\
\langle\phi_{\mathbf{k}_1}\phi_{\mathbf{k}_2}\phi_{\mathbf{k}_3}\phi_{\mathbf{k}41}\rangle_0 &= \langle\phi_{\mathbf{k}_1}\phi_{\mathbf{k}_2}\rangle_0\ \langle\phi_{\mathbf{k}_3}\phi_{\mathbf{k}_4}\rangle_0 + \langle\phi_{\mathbf{k}_1}\phi_{\mathbf{k}_3}\rangle_0\ \langle\phi_{\mathbf{k}_2}\phi_{\mathbf{k}_4}\rangle_0 + \langle\phi_{\mathbf{k}_1}\phi_{\mathbf{k}_4}\rangle_0\ \langle\phi_{\mathbf{k}_2}\phi_{\mathbf{k}_3}\rangle_0
\end{aligned}$$

and in general we find

$$\Big\langle\prod_{i=1}^{n}\phi_{\mathbf{k}_i}\Big\rangle_0 = \begin{cases} 0 \text{ if } n \text{ is odd} \\ \text{sum over all pairwise combinations if } n \text{ is even} \end{cases}$$

4.2 Perturbative Methods

Now we can extend the RG methods and scaling relations from section 3 to compute critical exponents in the Landau-Ginzburg theory. We want to find the RG flow starting from

$$\begin{aligned}
F_{\Lambda_0}[\phi] &= \frac{1}{2}\int\frac{d\mathbf{k}}{(2\pi)^d}(a_0 + \mathbf{k}^2)|\phi_{\mathbf{k}}|^2 + g\int\frac{d\mathbf{k}_1\ d\mathbf{k}_2\ d\mathbf{k}_3}{(2\pi)^{3d}}\phi\mathbf{k}_1\cdot\phi_{\mathbf{k}_2}\ \phi_{\mathbf{k}_3}\cdot\phi_{-\mathbf{k}_1-\mathbf{k}_2-\mathbf{k}_3} \\
&= F_0[\phi^-] + F_0[\phi^+] + F_I[\phi^-,\phi^+].
\end{aligned}$$

we begin by integrating out the UV modes

$$Z_\Lambda = \int D\phi^- e^{-F_\Lambda[\phi^-]} = \int D\phi^- e^{-F_0[\phi^-]}\int D\phi^+ e^{-F_0[\phi^+]}e^{-F_I[\phi^-,\phi^+]} = \int D\phi^- e^{-F_0[\phi^-]}\langle e^{-F_I[\phi^-,\phi^+]}\rangle_+$$

where the average in the last expression is over UV modes ϕ^+. We find

$$F_\Lambda[\phi^-] = F_0[\phi^-] - \log\langle e^{-F_I[\phi^-,\phi^+]}\rangle_+$$

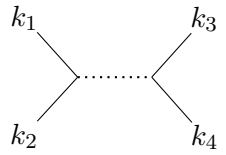
using that $\log(1+x) = x - \frac{1}{2}x^2$ and $e^x = 1 + x + \frac{1}{2}x^2 + ...$ we get $\log\langle e^{-F_I}\rangle = \log\langle 1 - F_I + \frac{1}{2}F_I^2 + ...\rangle = -\langle F_I\rangle + \frac{1}{2}\langle F_I^2\rangle - \frac{1}{2}\langle F_I\rangle^2 + ...$ and thus

$$F_\Lambda[\phi^-] = F_0[\phi^-] + \langle F_I\rangle_+ - \frac{1}{2}[\langle F_I^2\rangle_+ - \langle F_I\rangle_+^2] + ... = F_0[\phi^-] - \sum_{n=0}^{\infty}\frac{(-1)^n}{n!}\langle F_I^n\rangle_+^c$$

where $\langle\ \rangle_+^c$ denotes a connected expectation value.

The computation of expectation values can be facilitated by the use of diagrammatic methods to keep track of all possible pairwise averages ("contractions") due to Wick's theorem. Suppose we wish to compute the term of order n in F_I.

(i) Draw n vertices, where each vertex is four points joined by legs, corresponding to the factor of $g\phi_{\mathbf{k}_1}\cdot\phi_{\mathbf{k}_2}\ \phi_{\mathbf{k}_3}\cdot\phi_{-\mathbf{k}_1-\mathbf{k}_2-\mathbf{k}_3}$ appearing in the integral. We have to distinguish contractions involving dotted and non-dotted fields, so we represent each vertex by two points joined by a dotted line that represents the wavevector conserving delta function:



(ii) We draw all topologically distinct diagrams that can be obtained joining every two legs.

(iii) The value of each diagram is obtained by multiplying its components: each external line represents a factor of $\phi^-_{\mathbf{k}}$. Two joined lines represent a 2-point function evaluated at the points it connects, $\langle\phi^+\phi^+\rangle_+$, and a vertex contributes with a factor of g, a dot product between the appropriate fields, and a wavevector conserving delta-function $(2\pi)^d\delta^d(\mathbf{k}_1+\mathbf{k}_2+\mathbf{k}_3+\mathbf{k}_4)$.

(iv) We integrate over all the modes $\{\mathbf{k}_i\}$.

(v) Multiply by a numerical factor

$$\frac{(-1)^n}{n!}\times\ \text{number of different pairings leading to the same topology.}$$

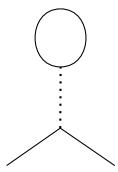
The first term comes from the sum expansion and the second term states that graphs with the same topology have the same value.

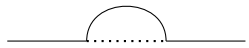
(vi) We sum over all such diagrams which are connected.

To exemplify we compute the 1st order (in g_0) correction to the free energy:

$$\langle F_I[\phi^+,\phi^-]\rangle_+ = \langle g_0\delta_{a_1a_2}\delta_{b_1b_2}\int\prod_{i=1}^{4}\Big(\frac{d\mathbf{k}_i}{(2\pi)^d}(\phi^{a_i,+}_{\mathbf{k}_i}+\phi^{a_i,-}_{\mathbf{k}_i})\Big)(2\pi)^d\delta^d\Big(\sum_{i=1}^{4}\mathbf{k}_i\Big)\rangle_+.$$

Terms involving and odd number of ϕ^+ vanish and the single term involving four ϕ^+ contributes with a constant to the free energy. The only non-trivial contributions of $\langle F_I\rangle_+$ to F_Λ are those proportional to $(\phi^-)^2$





$$
\begin{aligned}
\langle F_I\rangle_+ =& 2g_0 \int \prod_{i=1}^{4}\Big(\frac{d\mathbf{k}_i}{(2\pi)^d}\Big)\phi^-_{\mathbf{k}_1}\cdot\phi^-_{\mathbf{k}_2}\langle\phi^+_{\mathbf{k}_3}\cdot\phi^+_{\mathbf{k}_4}\rangle_+(2\pi)^d\delta^d\Big(\sum_{i=1}^{4}\mathbf{k}_i\Big)\\
&+4g_0\int\prod_{i=1}^{4}\Big(\frac{d\mathbf{k}_i}{(2\pi)^d}\Big)\phi^-_{\mathbf{k}_1}\cdot\langle\phi^+_{\mathbf{k}_3}\phi^+_{\mathbf{k}_4}\rangle_+\cdot\phi^-_{\mathbf{k}_2}(2\pi)^d\delta^d\Big(\sum_{i=1}^{4}\mathbf{k}_i\Big)\\
=&(2N+4)g_0\int\prod_{i=1}^{4}\Big(\frac{d\mathbf{k}_i}{(2\pi)^d}\Big)\phi^-_{\mathbf{k}_1}\cdot\phi^-_{\mathbf{k}_2}(2\pi)^d\frac{\delta^d(\mathbf{k}_3+\mathbf{k}_4)\delta^d(\sum_{i=1}^4\mathbf{k}_i)}{(a_0+\mathbf{k}^2)}\\
=&2(N+2)g_0\int\frac{d\mathbf{k}}{(2\pi)^d}\phi^-_{\mathbf{k}}\cdot\phi^-_{-\mathbf{k}}\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{1}{(a_0+\mathbf{q}^2)}
\end{aligned}
$$

so this term contributes to the $\frac{a}{2}\phi^2$ term in F_0:

$$
a' = a_0 + 4(N+2)g_0\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{1}{(a_0+\mathbf{q}^2)}
$$

The 2nd and 3rd steps of RG give, to 1st order in g_0:

$$
\begin{aligned}
a(\zeta) &= \zeta^2\left(a_0 + 4(N+2)g_0\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{1}{(a_0+\mathbf{q}^2)}\right)\\
g(\zeta) &= \zeta^{4-d}g_0
\end{aligned}
$$

So the ϕ^4 coupling induces a ϕ^2 coupling under the RG flow. Now the critical point $a(\zeta)=0$ is not equivalent to $a_0=0$ anymore, but to some other value proportional to g_0 (see figure 5). To hit the critical point, we must "fine tune" the temperature a_0 to cancel precisely the new terms that are generated by the RG flow.

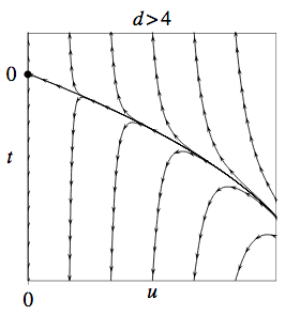


Figure 5: RG flow to first order in g. Notice that the quartic coupling shifts the transition temperature. In this figure, t and u correspond to a, g.

We now compute the 1st order (in g_0) correction to g_0. For that we need to go to order g_0^2 in the perturbative expansion:

$$
-\frac{1}{2}\langle F_I^2\rangle^c_+ = g_0^2\ \delta_{a_1a_2}\delta_{a_3a_4}\delta_{a_5a_6}\delta_{a_7a_8}\int\prod_{i=1}^{8}\langle\Big(\frac{d\mathbf{k}_i}{(2\pi)^d}(\phi^{a_i,+}_{\mathbf{k}_i}+\phi^{a_i,-}_{\mathbf{k}_i})\Big)\rangle^c_+\ (2\pi)^{2d}\delta^d\Big(\sum_{i=1}^{4}\mathbf{k}_i\Big)\delta^d\Big(\sum_{i=5}^{8}\mathbf{k}_i\Big)
$$

Non-trivial connected diagrams are given in table 3. The first diagram in the first row generates ϕ^6 terms. The second and third vanish because wavevector conservation on the rightmost vertex does not allow a situation where an IR mode goes in and an UV mode comes out.

The diagrams in the second and third rows are order g_0^2 contributions to a_0 so we are not interested in them here. The last two diagrams in the third row, however, have an interesting property: they

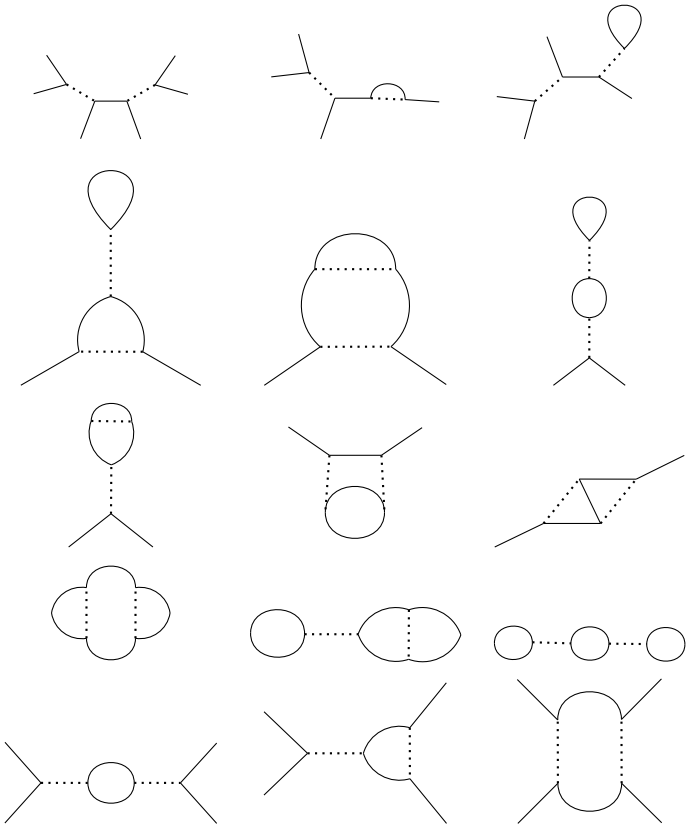
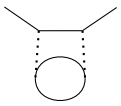


Table 3: Second order contributions to the free energy

contribute to the kinetic term, inducing a field renormalization in step 3 of the RG. This means that the field acquires an anomalous dimension $\eta \neq 0$. For instance, consider the first of these diagrams (ignoring the numerical factor):



$$\sim -\frac{g_0^2}{2}\int \frac{d\mathbf{k}d\mathbf{q}_1}{(2\pi)^2 d}\phi^-_{\mathbf{k}}\cdot\langle\phi^+_{\mathbf{q}_1}\phi^+_{-\mathbf{q}_1}\rangle_+\cdot\phi^-_{-\mathbf{k}}\int\frac{d\mathbf{q}_2}{(2\pi)^d}\langle\phi^{a,+}_{\mathbf{q}_2}\phi^{b,+}_{-\mathbf{q}_2}\rangle_+\langle\phi^{a,+}_{\mathbf{k}+\mathbf{q}_1+\mathbf{q}_2}\phi^{b,+}_{-\mathbf{k}-\mathbf{q}_1-\mathbf{q}_2}\rangle_+$$
$$= -\frac{g_0^2}{2}N\int\frac{d\mathbf{k}}{(2\pi)^d}\phi^-_{\mathbf{k}}\cdot\phi^-_{-\mathbf{k}}\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}_1 d\mathbf{q}_2}{(2\pi)^{2d}}\frac{1}{(a_0+\mathbf{q}_1)^2}\frac{1}{(a_0+\mathbf{q}_2^2)}\frac{1}{(a_0+(\mathbf{k}+\mathbf{q}_1+\mathbf{q}_2)^2)}.$$

Expanding the integrand in $\mathbf{k}$, the linear term gives an odd integral that vanishes, and we get to second order:

$$-\frac{g_0^2}{2}N\int\frac{d\mathbf{k}}{(2\pi)^d}\phi_{\mathbf{k}}\cdot\phi_{-\mathbf{k}}\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}_1 d\mathbf{q}_2}{(2\pi)^{2d}}\frac{1}{(a_0+\mathbf{q}_1)^2}\frac{1}{(a_0+\mathbf{q}_2^2)}\left\{\frac{1}{(a_0+(\mathbf{q}_1+\mathbf{q}_2)^2)}+\frac{\mathbf{k}^2}{2}\frac{6(\mathbf{q}_1+\mathbf{q}_2)^2-2a_0}{(a_0+(\mathbf{q}_1+\mathbf{q}_2)^2)^3}\right\}.$$

The first term in the brackets is a second order contribution to a_0. The second term, however, contributes to the kinetic term $\mathbf{k}^2|\phi_{\mathbf{k}}|^2$, such that the field renormalization is

$$\phi_{\mathbf{k}} = \frac{1}{\sqrt{1-Ng_0^2A}}\phi'_{\mathbf{k}'}$$

where

$$A = \int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}_1 d\mathbf{q}_2}{(2\pi)^{2d}}\frac{1}{(a_0+\mathbf{q}_1)^2}\frac{1}{(a_0+\mathbf{q}_2^2)}\frac{3(\mathbf{q}_1+\mathbf{q}_2)^2-a_0}{(a_0+(\mathbf{q}_1+\mathbf{q}_2)^2)^3}.$$

All diagrams in the fourth row contribute with a constant to the free energy. The last diagrams, proportional to $(\phi^-)^4$ are the interesting ones. These are the first order corrections to g_0 (which are of order g_0^2). The first diagram in the last row is

$$-\frac{1}{2}2^3\ Ng_0^2\int\prod_{i=1}^{4}\Big(\frac{d\mathbf{k}_i}{(2\pi)^d}\phi^{a_i,-}_{\mathbf{k}_i}\Big)(2\pi)^d\delta^d\Big(\sum_{i=1}^{4}\mathbf{k}_i\Big)\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{1}{(a_0+\mathbf{q}^2)}\frac{1}{(a_0+(\mathbf{q}-\mathbf{k}_1-\mathbf{k}_2)^2)}$$

where the numerical factor of 2^3 represents the number of possible pairings: 2 choices of legs for each vertex (2×2) multiplied by 2 possible pairings between them. The factor of N comes from $\delta_{ab}\delta_{ba}$. Expanding in $\mathbf{k}_1,\mathbf{k}_2$, we get terms that contribute to quartic gradient terms (which are irrelevant). Retaining only to order 0, we find the quartic coupling contribution:

$$-4Ng_0^2\int\prod_{i=1}^{4}\Big(\frac{d\mathbf{k}_i}{(2\pi)^d}\phi^{a_i,-}_{\mathbf{k}_i}\Big)(2\pi)^d\delta^d\Big(\sum_{i=1}^{4}\mathbf{k}_i\Big)\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{1}{(a_0+\mathbf{q}^2)^2}$$

The other diagrams are very similar, the only difference is the lack of factor of N and the numerical factor is $\frac{1}{2}4\times 4\times 2$ for the second diagram (4 ways to choose one leg from each vertex and 2 ways to choose one more leg) and $\frac{1}{2}2\times 2\times 2\times 2$ for the third (one choice of legs from each of the four points in the two vertices). Hence to order g_0^2 we get the first order contribution to g_0:

$$g_0' = g_0-(4N+32)g_0^2\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{1}{(a_0+\mathbf{q}^2)^2}$$

In d=4, ϕ^4 seemed to be marginal, but we see now that the 1st step of RG decreases the value of g_0, making it marginally irrelevant. That means that the theory is similar in $d=4$ to the $d>4$ case when we take perturbative effects into account - it flows back to the GFP.

To summarise, under RG we find

$$\begin{cases} a(\zeta)=\zeta^2(a_0+\alpha g_0)\\ g(\zeta)=\zeta^{4-d}(g_0-\beta g_0^2)\end{cases}$$

where

$$\alpha = 4(N+2)\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{1}{(a_0+\mathbf{q}^2)} = 4(N+2)\ \Omega_{d-1}\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{dq}{(2\pi)^d}\frac{q^{d-1}}{(a_0+q^2)}$$

$$\beta = 4(N+8)\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{1}{(a_0+\mathbf{q}^2)^2} = 4(N+8)\ \Omega_{d-1}\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{dq}{(2\pi)^d}\frac{q^{d-1}}{(a_0+\mathbf{q}^2)^2}$$

where $\Omega_{d-1}=\frac{2\pi^{d/2}}{\Gamma(d/2)}$ is the volume of the boundary of an unit sphere S^{d-1}. For instance, for a 1-sphere (a circle) $\Omega_1=2\pi$ (its circumference) and for a 2-sphere (a sphere) $\Omega_2=4\pi$ (its area).

It is common to convert these relations to differential equations by setting $\zeta=e^s$. Near $\zeta=1$ (i.e. $s=0$ or $\Lambda=\Lambda_0$) we obtain [1]

$$\frac{da}{ds} = 2a+g\ \frac{4(N+2)\Omega_{d-1}}{(2\pi)^d}\frac{\Lambda^d}{a+\Lambda^2}$$

$$\frac{dg}{ds} = (4-d)g-g^2\ \frac{4(N+8)\Omega_{d-1}}{(2\pi)^d}\frac{\Lambda^d}{(a+\Lambda^2)^2}$$

[1] We differentiate the integrals using

$$\int_{\Lambda/\zeta}^{\Lambda}dqf(q)\approx(\Lambda-\Lambda e^{-s})f(\Lambda)\approx\Lambda f(\Lambda)s+O(s^2).$$

or, specialising to the $d = 4$ case ($\Omega_3 = 2\pi^2$):

$$\frac{da}{ds} = 2a + g\,\frac{N+2}{2\pi^2}\frac{\Lambda^4}{a+\Lambda^2}$$
$$\frac{dg}{ds} = -g^2\,\frac{N+8}{2\pi^2}\frac{\Lambda^4}{(a+\Lambda^2)^2}$$

note that this holds only for s close to 0, where $a \approx a_0$ and $g \approx g_0$. The quantity on the right of $\frac{dg_i}{ds}$ is generally known as the beta function.

4.3 ϵ-expansion

We saw that for $d < 4$ the quartic coupling is relevant, which means that g is not a small parameter and perturbation theory is not valid, so it is not clear where the theory flows to. There is a trick to deal with this situation known as the ϵ-expansion, which involves considering out theory in dimension $d = 4 - \epsilon$. The equations in the previous section become, for small ϵ:

$$\frac{da}{ds} = 2a + \tilde{g}\,\frac{N+2}{2\pi^2}\frac{\Lambda^4}{a+\Lambda^2} \tag{8}$$
$$\frac{d\tilde{g}}{ds} = \epsilon\tilde{g} - \tilde{g}^2\,\frac{N+8}{2\pi^2}\frac{\Lambda^4}{(a+\Lambda^2)^2}$$

where we defined a dimensionless coupling $\tilde{g} = \Lambda^{-\epsilon}g$. We immediately see that these equations have two fixed points, the GFP $(a, \tilde{g}) = (0, 0)$ and the so-called Wilson-Fisher fixed point given by

$$a_* = -\tilde{g}_*\,\frac{N+2}{4\pi^2}\frac{\Lambda^4}{a_*+\Lambda^2}$$
$$\tilde{g}_* = \epsilon\,\frac{2\pi^2}{N+8}\frac{(a_*+\Lambda^2)^2}{\Lambda^4}.$$

Plugging the second expression in the first, we find that to leading order in ϵ

$$a_* = -\tilde{g}_*\,\frac{N+2}{4\pi^2}\Lambda^2 = -\frac{N+2}{2(N+8)}\Lambda^2\epsilon$$
$$\tilde{g}_* = \frac{2\pi^2}{N+8}\epsilon$$

so we see that $(a_*, g_*) \approx O(\epsilon) \ll 0$ and that justifies using the equations that are valid only for small s.

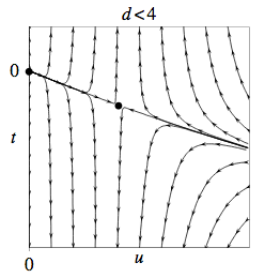


Figure 6: RG flow to first order in ϵ near the Wilson-Fisher fixed point. Again, t and u correspond to a, g.

We want now to calculate critical exponents near the Wilson-Fisher fixed point and compare them to experimental results for $d < 4$. For that we need to analyse the flow near that point, and this is done linearising the equations (8) setting $a(\zeta) = a_* + \delta a$ and $\tilde{g}(\zeta) = \tilde{g}_* + \delta\tilde{g}$:

$$\frac{d}{ds}\begin{pmatrix}\delta a\\ \delta\tilde{g}\end{pmatrix} = \begin{pmatrix}(2 - \frac{N+2}{N+8}\epsilon) & \frac{N+2}{2\pi^2}\Lambda^2(1 + \frac{N+2}{2(N+8)}\epsilon)\\ 0 & -\epsilon\end{pmatrix}\begin{pmatrix}\delta a\\ \delta\tilde{g}\end{pmatrix}$$

As the matrix is upper triangular (to order ϵ), its eigenvalues are just the diagonal entries, so

$$\delta a(\zeta) = \zeta^{2-\frac{N+2}{N+8}\epsilon}\delta a$$
$$\delta\tilde{g}(\zeta) = \zeta^{-\epsilon}\delta\tilde{g}$$

and therefore, in $d = 4-\epsilon$ dimensions, the Wilson-Fisher fixed point has one relevant and one irrelevant direction (see figure 6). This means that, provided we tune a accordingly, following the RG flow we will hit the Wilson-Fisher fixed point. For $a > a_*$ we find a disordered phase, and for $a < a_*$ an ordered phase with Goldstone modes.

	$d \geq 4$ (MF)	$\epsilon = 1$	$d = 3$
α	0	0.17	0.11
β	1/2	0.33	0.33
γ	1	1.17	1.24
δ	3	4	4.79
ν	1/2	0.58	0.63
η	0	0	0.036

Table 4: Critical exponents for $N = 1$ in $d \geq 4, \epsilon = 1$ and experimental results for $d = 3$.

Now we can compute critical exponents.

$$\nu = \Delta_a^{-1} = \frac{1}{2 - \frac{N+2}{N+8}\epsilon} \approx \frac{1}{2} + \frac{N+2}{4(N+8)}\epsilon.$$

We can compute η using our expression for the field renormalization (which is, remember, of order g_0^2): after the 3 steps of RG the kinetic term becomes

$$\frac{1}{2}\int \frac{d\mathbf{k}'}{(2\pi)^d}\zeta^{-d}\mathbf{k}'^2\zeta^{-2}|\phi_{\mathbf{k}'}|^2\zeta^{2d-2\Delta_\phi}(1 - Ng_o^2 A)$$

and therefore to renormalize the field we must have

$$-d - 2 + 2d - 2\Delta_\phi + \log_\zeta(1 - Ng^2 A) = 0$$

where $\Delta_\phi = \frac{d-2+\eta}{2}$. Expanding the log to first order in g and using that $g_* \sim \epsilon$, we find $\eta \sim \epsilon^2$ so it can be neglected.

Using the scaling relations from section 3.1, we find the exponents

$$\alpha = \frac{4-N}{2(N+8)}\epsilon \qquad \beta = \frac{1}{2} - \frac{3}{2(N+8)}\epsilon \qquad \gamma = 1 + \frac{N+2}{2(N+8)}\epsilon \qquad \delta = 3 + \epsilon$$

These results, of course, are supposed to be valid for very small ϵ. If we, however, just plug in $\epsilon = 1$, it turns out that the exponents match experimental results very well, as shown in table 4. These exponents characterise the $O(N)$ universality class, also known as Ising or para-ferromagnetic universality class.

5 Constrained Free Energies

In section 3, we showed how all critical exponents can be related to ν, which expresses how the correlation length diverges near a critical point, and that defines a universality class. It was thought that ν depended only on the symmetries, that is models with the same symmetries would present the same critical exponents. It has become apparent, though, that this is not true.

To illustrate this, we consider in this section an $O(2)$ model in $d = 2$ dimensions with an additional constraint [1]. We will see that this model lies in a different universality class as a consequence of the constraint. We take as our model a liquid crystal, composed of rod-shaped molecules. They are described by a coarse-grained vector field $\mathbf{n}(\mathbf{r})$ which represents the direction that molecules tend to align themselves at position $\mathbf{r}$, either parallely or antiparallely. We take it to be an unit vector, $\mathbf{n} \cdot \mathbf{n} = 1$

These molecules need not be straight - they can be bent ("banana" shaped) or triangular ("pizza" shaped). It is observed that as we increase the density or decrease the temperature, these systems undergo a phase transition to a spontaneously bent (in the case of banana) or splayed (in the case of pizza) phase, as shown in figure 7. We refer to these phases in general as splay-bend (SB).

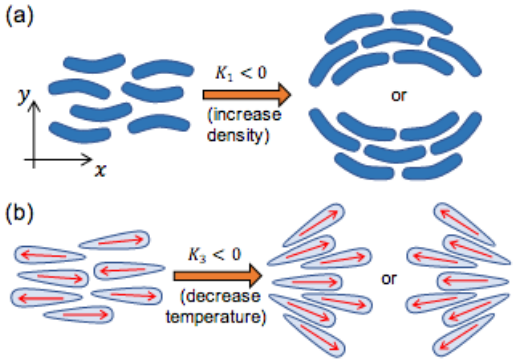


Figure 7: (a) Banana-shaped and (b) pizza-shaped molecules going through a phase transition. Reproduced from [1].

The free energy under consideration is

$$F[\mathbf{n}] = \int d\mathbf{r} f(\mathbf{n}) = \int d\mathbf{r} \Big\{ \frac{K_1}{2} (\nabla \cdot \mathbf{n})^2 + \frac{K_3}{2} |\mathbf{n} \times (\nabla \times \mathbf{n})|^2 + \frac{C}{2} |\nabla^2 \mathbf{n}|^2 \Big\}.$$

The first term $\nabla \cdot n > 0$ describes the divergence of the vector field (splay), and K_1 can be negative to favour spontaneous splay. The second term describes bending - the vector product of $\nabla \times \mathbf{n}$ and $\mathbf{n}$ is proportional to the bending direction, and K_3 positive or negative can penalise or favour bent configurations. Finally, the last term with $C > 0$ ensures that the free energy is stable (see [1]).

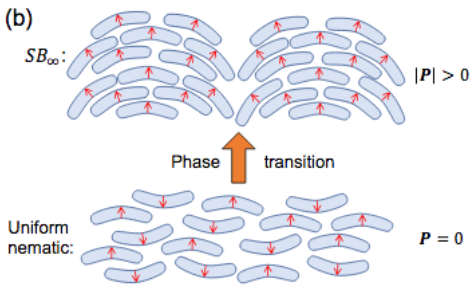


Figure 8: Nematic-SB_∞ transition. $\mathbf{P}$ stands for the polarisation field. Reproduced from [1].

We will show that this system displays a phase transition from an uniform nematic phase $\mathbf{n} =$ const. to a phase we shall call SB_∞, which is schematised in figure 8 for a polar banana-shaped molecule. We note that the polarisation field is in the bending direction $\mathbf{n} \times (\nabla \times \mathbf{n})$, such that the nematic phase has 0 net polarisation and non zero polarisation in the SB_∞ phase.

We start our analysis by taking $\mathbf{n}(\mathbf{r}) = (\cos\theta(\mathbf{r}), \sin\theta(\mathbf{r}))$ and deriving the following identities:

$$\begin{aligned}
(\nabla\cdot\mathbf{n})^2 =&(-\sin\theta\partial_x\theta + \cos\theta\partial_y\theta)^2 = \sin^2\theta(\partial_x\theta)^2 - \sin 2\theta\partial_x\theta\partial_y\theta + \cos^2\theta(\partial_y\theta)^2\\
=&(\nabla\theta)^2 - \cos^2\theta(\partial_x\theta)^2 - \sin^2\theta(\partial_y\theta)^2 - \sin 2\theta\partial_x\theta\partial_y\theta\\
=&(\nabla\theta)^2 - \frac{\cos 2\theta}{2}(\partial_x\theta)^2 - \frac{1}{2}(\partial_x\theta)^2 + \frac{\cos 2\theta}{2}(\partial_y\theta)^2 - \frac{1}{2}(\partial_y\theta)^2 - \sin 2\theta\partial_x\theta\partial_y\theta\\
=&\frac{1}{2}(\nabla\theta)^2 - \frac{1}{2}[(\partial_x\theta)^2 - (\partial_y\theta)^2]\cos 2\theta - \sin 2\theta\partial_x\theta\partial_y\theta
\end{aligned}$$

$$\begin{aligned}
|\mathbf{n}\times\nabla\times\mathbf{n}|^2 =&|\mathbf{n}\times(\cos\theta\partial_x\theta + \sin\theta\partial_y\theta)\mathbf{z}|^2\\
=&|(\sin\theta\cos\theta\partial_x\theta + \sin^2\theta\partial_y\theta, -\cos^2\theta\partial_x\theta - \sin\theta\cos\theta\partial_y\theta)|^2\\
=&\sin^2\theta\cos^2\theta(\partial_x\theta)^2 + 2\sin^3\theta\cos\theta\partial_x\theta\partial_y\theta + \sin^4\theta(\partial_y\theta)^2\\
&+\sin^2\theta\cos^2\theta(\partial_y\theta)^2 + 2\sin\theta\cos^3\theta\partial_x\theta\partial_y\theta + \cos^4\theta(\partial_x\theta)^2\\
=&\cos^2\theta(\partial_x\theta)^2 + \sin^2\theta(\partial_y\theta)^2 + 2\sin\theta\cos\theta\partial_x\theta\partial_y\theta\\
=&(\nabla\theta)^2 - \sin^2\theta(\partial_x\theta)^2 - \cos^2\theta(\partial_y\theta)^2 + \sin 2\theta\partial_x\theta\partial_y\theta\\
=&(\nabla\theta)^2 + \frac{\cos 2\theta}{2}(\partial_x\theta)^2 - \frac{1}{2}(\partial_x\theta)^2 - \frac{\cos 2\theta}{2}(\partial_y\theta)^2 - \frac{1}{2}(\partial_y\theta)^2 + \sin 2\theta\partial_x\theta\partial_y\theta\\
=&\frac{1}{2}(\nabla\theta)^2 + \frac{1}{2}[(\partial_x\theta)^2 - (\partial_y\theta)^2]\cos 2\theta + \sin 2\theta\partial_x\theta\partial_y\theta
\end{aligned}$$

$$\begin{aligned}
|\nabla^2\mathbf{n}|^2 =&(\partial_x^2\mathbf{n})^2 + (\partial_y^2\mathbf{n})^2 = \sum_{i=x,y}(\partial_i(-\sin\theta\partial_i\theta, \cos\theta\partial_i\theta))^2\\
=&\sum_{i=x,y}(-\cos\theta(\partial_i\theta)^2 - \sin\theta\partial_i^2\theta, -\sin\theta(\partial_i\theta)^2 + \cos\theta\partial_i^2\theta)^2 = \sum_{i=x,y}(\partial_i\theta)^4 + (\partial_i^2\theta)^2\\
=&(\nabla\theta)^4 + (\nabla^2\theta)^2.
\end{aligned}$$

We can use these identities to rewrite the free energy as

$$\begin{aligned}
F[\theta] = \int d\mathbf{r}\Big(&\frac{K_1+K_3}{4}|\nabla\theta|^2 + \frac{C}{2}(\nabla^2\theta)^2 + \frac{C}{2}|\nabla\theta|^4 - \frac{K_1-K_3}{4}[(\partial_x\theta)^2 - (\partial_y\theta)^2]\cos(2\theta)\\
&-\frac{K_1-K_3}{2}\partial_x\theta\partial_y\theta\sin(2\theta)\Big).
\end{aligned}$$

We look for mean field solutions in some different regimes. For $K_1 > 0$ and $K_3 > 0$, any fluctuation is penalised - this is the nematic phase. We take the spontaneously broken direction to be $\mathbf{n} = (1, 0)$ such that this phase is characterised by $\theta = 0$.

As we decrease K_1 to a negative value, favouring splay over bend, we expect the mean field to oscillate in the direction perpendicular to $(1, 0)$, as shown in figure 9. We refer to this phase as $SB_\perp$ and it is formed by pizza-shaped molecules. We assume a mean field $\theta(\mathbf{r}) = \theta_0\cos(k_0 y)$ and substitute into the free energy density $f(\theta)$, obtaining

$$\begin{aligned}
f(\theta) =&\frac{K_1+K_3}{4}\theta_0^2k_0^2\sin^2(k_0y) + \frac{C}{2}\theta_0^2k_0^4\cos^2(k_0y)\\
&+\frac{K_1-K_3}{4}\theta_0^2k_0^2\sin^2(k_0y)\cos(2\theta_0\cos(k_0y)) + O(\theta_0^4k_0^4)
\end{aligned}$$

Expanding the cos(cos) term and averaging the expression over one wavelength $\lambda_0 = \frac{2\pi}{k_0}$ we find

$$\bar{f} = \frac{1}{4}K_1\theta_0^2k_0^2 + \frac{1}{16}(K_3 - K_1)\theta_0^4k_0^2 + \frac{C}{4}\theta_0^2k_0^4,$$

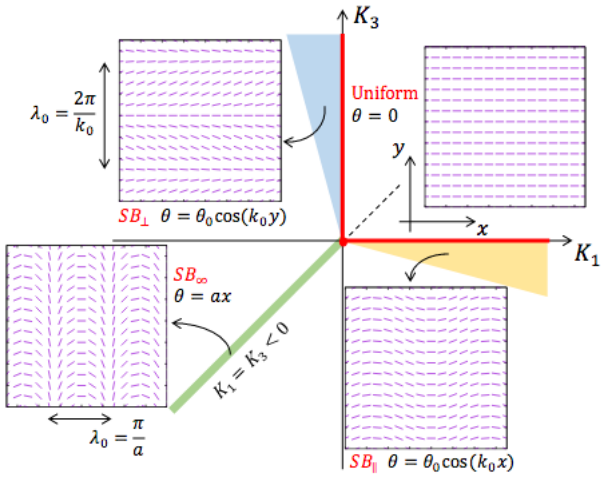


Figure 9: Phase diagram for banana and pizza shaped molecules. Reproduced from [1].

where I used that $\langle \sin^2(k_0 y)\rangle_{\lambda_0} = \frac{1}{2}$ and

$$\langle \sin^2(k_0 y)\cos^2(k_0 y)\rangle_{\lambda_0} = \frac{1}{4\lambda_0}\int_0^{\lambda_0} \sin^2(k_0 y)\cos^2(k_0 y)dy = \frac{1}{4\lambda_0}\int_0^{\lambda_0} \sin^2(2k_0 y)dy = \frac{1}{8}. \tag{9}$$

The mean field exponents can then be obtained by minimising the free energy over θ_0, k_0:

$$\theta_0 = \sqrt{\frac{4}{3}\frac{-K_1}{K_3 - K_1}}$$
$$k_0 = \sqrt{\frac{-K_1}{3C}}$$

and we see that the order parameter changes as $\theta_0 \sim |K_1|^\beta$ with $\beta = \frac{1}{2}$. Note that the approximations that were made are only valid if θ_0, k_0 are small, i.e. K_1 is negative small and $K_3 \gg |K_1|$, which corresponds to the blue shaded region in figure 9.

The mean field in the $SB_\parallel$ phase can be calculated in a similar way and it corresponds to the case $K_3 < 0, K_1 \gg |K_3|$. It also oscillates but now in a direction parallel to the spontaneously broken director, $\theta(\mathbf{r}) = \theta_0 \cos(k_0 x)$. This can describe banana-shaped molecules.

Most of the other regions in the phase diagram are not known analytically, except for SB_∞. This phase describes the behaviour of polar banana-shaped molecules and corresponds to $K_1 = K_3 = K < 0$. Its free energy is

$$F[\theta] = \int d\mathbf{r}\Big(\frac{K}{2}|\nabla\theta|^2 + \frac{C}{2}(\nabla^2\theta)^2 + \frac{C}{2}|\nabla\theta|^4\Big).$$

Defining a vector field $m(\mathbf{r}) = \nabla\theta(\mathbf{r})$, we can rewrite this as

$$F[m] = \int d\mathbf{r}\Big(\frac{C}{2}(\nabla\cdot m)^2 + \frac{K}{2}|m|^2 + \frac{C}{2}|m|^4\Big).$$

This is identical to an O(N=d) model (with d the number of dimensions, 2 in the present case) subject to the constraint $\nabla \times m = 0$. We will show that this model does not lie in the Ising/ferromagnetic universality class from section 4.3.

As we already know, the mean field solution for this model is $\bar{m} = (a, b)$ such that $a^2 + b^2 = \frac{-K}{2C}$. We can integrate this to find

$$\theta(\mathbf{r}) = ax + by.$$

We set the spontaneously broken direction along the $\hat{x}$-axis and let the mean field be

$$\bar{m} = \Big(\sqrt{\frac{-K}{2C}}, 0\Big).$$

Borrowing the results from section 2.4, around this mean field we have

$$F[m] = \int d\mathbf{r}\Big\{\frac{C}{2}(\nabla \cdot m)^2 - Km_x^2\Big\}$$

subject to $\nabla \times m = 0$.

It looks like m_y became a Goldstone mode. However, this is not the case: the constraint $\nabla \times m = \partial_x m^y - \partial_y m^x = 0$ forces the massless mode to be dependent on the massive one. Therefore we find no Goldstone modes. This seems to contradict the Mermin-Wagner theorem, but it can be understood as a Higgs-like mechanism: the constraint can be included in the free energy as a Lagrange multiplier $\lambda(\mathbf{r})\nabla \times m$. We can think of $\lambda(\mathbf{r})$ as a "Higgs" field that gives mass to the Goldstone mode such that there is no contradiction.

In Fourier space the free energy becomes

$$F[m] = \frac{1}{2}\int \frac{d\mathbf{k}}{(2\pi)^2}(Ck_ak_b - 2K\delta_{xa}\delta_{xb})m^a_{\mathbf{k}}m^b_{-\mathbf{k}}.$$

The constraint in Fourier space is $k_x m^y_{\mathbf{k}} - k_y m^x_{\mathbf{k}} = 0$ so that

$$F[m] = \frac{1}{2}\int \frac{d\mathbf{k}}{(2\pi)^2}(Ck_x^2 + 2Ck_y^2 + C\frac{k_y^4}{k_x^2} - 2K)|m^x_{\mathbf{k}}|^2.$$

and we find the correlation function

$$\langle m^x_{\mathbf{k}} m^x_{-\mathbf{k}}\rangle = -\frac{1}{K}\frac{1}{\xi^2(k_x^2 + 2k_y^2 + \frac{k_y^4}{k_x^2}) + 2}$$

where the correlation length was defined as $\xi^2 = -C/K$.

We therefore find a critical exponent of $\nu = \frac{1}{2}$, so at mean field level it is not possible to distinguish from the Ising universality class. Referring to section 4 we can improve the MF results using a perturbative approach. To perform the RG analysis we use the free energy in terms of θ, which we can write as:

$$\begin{aligned}
F[\theta] =& \int d\mathbf{r}\Big(\frac{K}{2}|\nabla\theta|^2 + \frac{C}{2}(\nabla^2\theta)^2 + L|\nabla\theta|^4\Big) \\
=& \frac{1}{2}\int \frac{d\mathbf{k}}{(2\pi)^d}(K\mathbf{k}^2 + C\mathbf{k}^4)|\theta_{\mathbf{k}}|^2 + L\int \frac{d\mathbf{k}_1 d\mathbf{k}_2 d\mathbf{k}_3 d\mathbf{k}_4}{(2\pi)^{4d}}\delta\Big(\sum_{i=1}^4 \mathbf{k}_i\Big)(\mathbf{k}_1\theta_{\mathbf{k}_1}\cdot\mathbf{k}_2\theta_{\mathbf{k}_2})(\mathbf{k}_3\theta_{\mathbf{k}_3}\cdot\mathbf{k}_4\theta_{\mathbf{k}_4}) \\
=& F_0[\theta] + F_I[\theta].
\end{aligned}$$

and at the end we can set $L = \frac{C}{2}$ again. The Gaussian correlation function is

$$\langle\theta_{\mathbf{q}}\theta_{\mathbf{q}'}\rangle_0 = \frac{\delta(\mathbf{q}+\mathbf{q}')}{K\mathbf{q}^2 + C\mathbf{q}^4}.$$

We begin by integrating out UV modes:

$$F_\Lambda[\theta^-] = F_0[\theta^-] + \langle F_I\rangle_+ - \frac{1}{2}\Big(\langle F_I^2\rangle_+ - \langle F_I\rangle_+^2\Big)$$

the non-trivial contributions to K to first order in L are

$$= 2L \int \prod_{i=1}^{4} \Big(\frac{d\mathbf{k}_i}{(2\pi)^d}\Big)(\mathbf{k}_1\cdot\mathbf{k}_2)(\mathbf{k}_3\cdot\mathbf{k}_4)\theta^-_{\mathbf{k}_1}\theta^-_{\mathbf{k}_2}\langle\theta^+_{\mathbf{k}_3}\theta^+_{\mathbf{k}_4}\rangle_+(2\pi)^d\delta^d(\sum_{i=1}^{4}\mathbf{k}_i)$$

$$= 2L \int \frac{d\mathbf{k}}{(2\pi)^d}\mathbf{k}^2|\theta_{\mathbf{k}}|^2 \int_{\frac{\Lambda}{\zeta}}^{\Lambda_0} \frac{d\mathbf{q}}{(2\pi)^d}\frac{q^2}{Kq^2+Cq^4}$$

$$= 4L \int \frac{d\mathbf{k}}{(2\pi)^d}k_a k_b|\theta_{\mathbf{k}}|^2 \int_{\frac{\Lambda}{\zeta}}^{\Lambda_0} \frac{d\mathbf{q}}{(2\pi)^d}\frac{q_a q_b}{Kq^2+Cq^4}$$

where each line represents a factor of $\mathbf{k}_i\theta_{\mathbf{k}_i}$. The last integral is isotropic: it has the same value (suppose A) if we rotate it with an O(d) matrix R. But $RAR^T = A \iff A = aI$. The constant a can be obtained by taking the trace:

$$\int \frac{d\mathbf{q}}{(2\pi)^d}\frac{q^2}{Kq^2+Cq^4} = a\times d$$

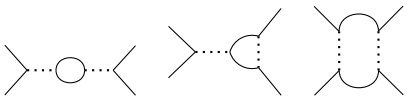
such that we get

$$= \frac{4L}{d}\int \frac{d\mathbf{k}}{(2\pi)^d}k^2|\theta_{\mathbf{k}}|^2 \int_{\frac{\Lambda}{\zeta}}^{\Lambda_0} \frac{d\mathbf{q}}{(2\pi)^d}\frac{1}{K+Cq^2}$$

and hence

$$K' = K + 4L\Big(1+\frac{2}{d}\Big)\int_{\frac{\Lambda}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{1}{K+Cq^2}.$$

To find the first order correction to L, we need to go to second order in the perturbative expansion. The non-trivial connected diagrams are:



For the first diagram we have

$$-\frac{8L^2}{2}\int\prod_{i=1}^{4}\Big(\frac{d\mathbf{k}_i}{(2\pi)^d}\Big)(2\pi)^d\delta^d(\sum_{i=1}^{4}\mathbf{k}_i)(\mathbf{k}_1\cdot\mathbf{k}_2)(\mathbf{k}_3\cdot\mathbf{k}_4)\theta_{\mathbf{k}_1}\theta_{\mathbf{k}_2}\theta_{\mathbf{k}_3}\theta_{\mathbf{k}_4}\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{q^4}{(Kq^2+Cq^4)^2}$$

$$= -4L^2\int\prod_{i=1}^{4}\Big(\frac{d\mathbf{k}_i}{(2\pi)^d}\Big)(2\pi)^d\delta^d(\sum_{i=1}^{4}\mathbf{k}_i)(\mathbf{k}_1\cdot\mathbf{k}_2)(\mathbf{k}_3\cdot\mathbf{k}_4)\theta_{\mathbf{k}_1}\theta_{\mathbf{k}_2}\theta_{\mathbf{k}_3}\theta_{\mathbf{k}_4}\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{1}{(K+Cq^2)^2}$$

where, as in section 4, we expanded the UV integral and ignored $O(\mathbf{k}_i)$ terms. For the second diagram:

$$-32\times\frac{L^2}{2}\int\prod_{i=1}^{4}\Big(\frac{d\mathbf{k}_i}{(2\pi)^d}\Big)(2\pi)^d\delta^d(\sum_{i=1}^{4}\mathbf{k}_i)(\mathbf{k}_1\cdot\mathbf{k}_2)k_{3,a}k_{4,b}\theta_{\mathbf{k}_1}\theta_{\mathbf{k}_2}\theta_{\mathbf{k}_3}\theta_{\mathbf{k}_4}\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{q^2q_aq_b}{(Kq^2+Cq^4)^2}$$

$$= -\frac{16}{d}L^2\int\prod_{i=1}^{4}\Big(\frac{d\mathbf{k}_i}{(2\pi)^d}\Big)(2\pi)^d\delta^d(\sum_{i=1}^{4}\mathbf{k}_i)(\mathbf{k}_1\cdot\mathbf{k}_2)(\mathbf{k}_3\cdot\mathbf{k}_4)\theta_{\mathbf{k}_1}\theta_{\mathbf{k}_2}\theta_{\mathbf{k}_3}\theta_{\mathbf{k}_4}\times\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{1}{(K+Cq^2)^2}$$

and finally, again ignoring $O(\mathbf{k}_i)$ terms in the integral

$$-32\times\frac{L^2}{2}\int\prod_{i=1}^{4}\Big(\frac{d\mathbf{k}_i}{(2\pi)^d}\Big)(2\pi)^d\delta^d(\sum_{i=1}^{4}\mathbf{k}_i)k_{1,a}k_{2,b}k_{3,c}k_{4,d}\theta_{\mathbf{k}_1}\theta_{\mathbf{k}_2}\theta_{\mathbf{k}_3}\theta_{\mathbf{k}_4}\times\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{q_aq_bq_cq_d}{(Kq^2+Cq^4)^2}$$

$$= -\frac{48}{d(d+2)}L^2\int\prod_{i=1}^{4}\Big(\frac{d\mathbf{k}_i}{(2\pi)^d}\Big)(2\pi)^d\delta^d(\sum_{i=1}^{4}\mathbf{k}_i)(\mathbf{k}_1\cdot\mathbf{k}_2)(\mathbf{k}_3\cdot\mathbf{k}_4)\theta_{\mathbf{k}_1}\theta_{\mathbf{k}_2}\theta_{\mathbf{k}_3}\theta_{\mathbf{k}_4}\times\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0}\frac{d\mathbf{q}}{(2\pi)^d}\frac{1}{(K+Cq^2)^2}$$

where it was used that if the tensor integral is isotropic and totally symmetric $R_i^a R_j^b R_k^c R_l^d A_{abcd} = A_{ijkl} \iff A_{abcd} = a(\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc})$ and taking the trace over (ab) and then (cd) we find

$$\int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0} \frac{d\mathbf{q}}{(2\pi)^d} \frac{q^4}{(Kq^2 + Cq^4)^2} = ad(d+2).$$

Therefore, putting everything together:

$$L' = L - 4L^2\Big(1 + \frac{4}{d} + \frac{12}{d(d+2)}\Big) \int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0} \frac{d\mathbf{q}}{(2\pi)^d} \frac{1}{(K + Cq^2)^2}.$$

Next we rescale momenta $\mathbf{k}' = \zeta\mathbf{k}$ and renormalize the field $\theta'_{\mathbf{k}'} = \zeta^{\Delta_\theta}\theta_{\mathbf{k}}$. To find Δ_θ, instead of requiring a canonical kinetic term we require that C remains invariant. This coupling does not get any corrections to first order in L so we find

$$\zeta^{-d-4}\zeta^{-2\Delta_\theta} C = C \to \Delta_\theta = \frac{d+4}{2}$$

and therefore

$$\begin{cases} K(\zeta) = \zeta^2\Big[K + 4L\Big(1 + \frac{2}{d}\Big) \int_{\frac{\Lambda}{\zeta}}^{\Lambda_0} \frac{d\mathbf{q}}{(2\pi)^d} \frac{1}{K+Cq^2}\Big] \\ L(\zeta) = \zeta^{4-d}\Bigg[L - 4L^2\Big(1 + \frac{4}{d} + \frac{12}{d(d+2)}\Big) \int_{\frac{\Lambda_0}{\zeta}}^{\Lambda_0} \frac{d\mathbf{q}}{(2\pi)^d} \frac{1}{(K+Cq^2)^2}\Bigg]. \end{cases}$$

setting $\zeta = e^s$ in analogy to section 4 we arrive at

$$\begin{cases} \frac{dK}{ds} = 2K + 4L\Big(1 + \frac{2}{d}\Big) \frac{\Omega_{d-1}}{(2\pi)^d} \frac{\Lambda^d}{K+C\Lambda^2} \\ \frac{dL}{ds} = (4-d)L - 4L^2\Big(1 + \frac{4}{d} + \frac{12}{d(d+2)}\Big) \frac{\Omega_{d-1}}{(2\pi)^d} \frac{\Lambda^d}{(K+C\Lambda^2)^2}. \end{cases}$$

For $d \geq 4$, we find a Gaussian fixed point at $(0,0)$ where K is relevant and L is irrelevant. Linearising around the GFP:

$$\frac{d}{ds}\begin{pmatrix} \delta K \\ \delta L \end{pmatrix} = \begin{pmatrix} 2 & 4\Big(1 + \frac{2}{d}\Big) \frac{\Omega_{d-1}}{(2\pi)^d} \frac{\Lambda^d}{K+C\Lambda^2} \\ 0 & 4-d \end{pmatrix} \begin{pmatrix} \delta K \\ \delta L \end{pmatrix}$$

so close to the GFP $\delta K(\zeta) = \zeta^2 \delta K$ such that $\nu = \Delta^{-1} = \frac{1}{2}$ as predicted by MFT. On the other hand, for $d < 4$, L becomes relevant and we can use the ϵ-expansion to find a non-trivial fixed point (K_*, L_*) (to first order in ϵ):

$$\begin{aligned} K_* &= -\frac{L}{4\pi^2}\Big(\frac{3}{2} + \frac{\epsilon}{8}\Big)\frac{\Lambda^2}{C} = -\frac{3}{10}\Lambda^2 C\epsilon \\ L_* &= \frac{4\pi^2}{5} C^2 \epsilon \end{aligned}$$

Linearising the equations around this fixed point, we find, retaining only to first order in ϵ:

$$\frac{d}{ds}\begin{pmatrix} \delta K \\ \delta L \end{pmatrix} = \begin{pmatrix} 2 - \frac{3}{5}\epsilon & \frac{3\Lambda^2}{4\pi^2 C} + \frac{23\Lambda^2}{80\pi^2}\epsilon \\ 0 & -\epsilon \end{pmatrix} \begin{pmatrix} \delta K \\ \delta L \end{pmatrix}$$

and therefore

$$\begin{aligned} \nu &= \Delta_K^{-1} = \frac{1}{2} + \frac{3}{20}\epsilon + O(\epsilon^2) \\ \alpha &= 2 - d\nu = \frac{\epsilon}{10} \\ \beta &= \frac{(d-2+\eta)\nu}{2} = \frac{1}{2} - \frac{\epsilon}{10} \\ \gamma &= \nu(2-\eta) = 1 + \frac{3}{10}\epsilon \\ \delta &= \frac{d+2-\eta}{d-2+\eta} = 3 + \epsilon. \end{aligned}$$

where we have used again that $\eta = 0$ to first order in L as there is no contribution to the field renormalisation. We can compare this to the formulae in the end of section 4.3 and check that we indeed have a different universality class.

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